

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING (IOE)

DIGITAL LOGIC

Course Code: ENEX 152 / EX 152 / CT 152

BE Computer (BCT) & BE Electronics, Information and Communication (BEI)

Comprehensive Master Examination Solutions

Complete step-by-step analytical solutions, excitation tables, full K-maps, state transition diagrams, timing waveforms, and publication-grade TikZ logic schematics for all exam sittings.

Academic Exam Papers Covered:

2083 Baishakh · 2082 Bhadra · 2082 Baishakh · 2081 Ashwin

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Chapter 1

2083 Baishakh Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)	Full Marks: 60
Programme: BCT, BEI	Pass Marks: 24
Year / Part: I / II	Time: 3 Hours
Subject: Digital Logic (ENEX 152)	Examination Type: Regular

1.0.1 Question 1: Gray Code to Binary Conversion [2 + 2 = 4 Marks]

Problem Statement (2083 Baishakh, Q1)

Convert the following gray codes to binary codes:

- (a) $(10011)_{\text{Gray}}$
- (b) $(110001)_{\text{Gray}}$

Conversion Principle and Algorithm: Let an n -bit Gray code word be $G = g_{n-1}g_{n-2} \dots g_1g_0$ and the corresponding binary code word be $B = b_{n-1}b_{n-2} \dots b_1b_0$.

1. The Most Significant Bit (MSB) of the binary code is identical to the MSB of the Gray code:

$$b_{n-1} = g_{n-1}$$

2. Each subsequent binary bit b_i (for $i = n - 2$ down to 0) is obtained by XORing the previously determined binary bit b_{i+1} with the current Gray code bit g_i :

$$b_i = b_{i+1} \oplus g_i \quad \text{for } i = n - 2, n - 3, \dots, 0$$

Part (a): Conversion of $(10011)_{\text{Gray}}$ (5-bit code) Given $g_4g_3g_2g_1g_0 = 10011_2$:

- $b_4 = g_4 = 1$
- $b_3 = b_4 \oplus g_3 = 1 \oplus 0 = 1$
- $b_2 = b_3 \oplus g_2 = 1 \oplus 0 = 1$
- $b_1 = b_2 \oplus g_1 = 1 \oplus 1 = 0$
- $b_0 = b_1 \oplus g_0 = 0 \oplus 1 = 1$

Thus, $(10011)_{\text{Gray}} = (11101)_2$.

Part (b): Conversion of $(110001)_{\text{Gray}}$ (6-bit code) Given $g_5g_4g_3g_2g_1g_0 = 110001_2$:

- $b_5 = g_5 = 1$
- $b_4 = b_5 \oplus g_4 = 1 \oplus 1 = 0$
- $b_3 = b_4 \oplus g_3 = 0 \oplus 0 = 0$
- $b_2 = b_3 \oplus g_2 = 0 \oplus 0 = 0$
- $b_1 = b_2 \oplus g_1 = 0 \oplus 0 = 0$
- $b_0 = b_1 \oplus g_0 = 0 \oplus 1 = 1$

Thus, $(110001)_{\text{Gray}} = (100001)_2$.

Final Verified Result

- (a) $(10011)_{\text{Gray}} = (\mathbf{11101})_2 = (29)_{10}$
 (b) $(110001)_{\text{Gray}} = (\mathbf{100001})_2 = (33)_{10}$

1.0.2 Question 2: Analog vs. Digital Signals & Excess-3 Code [1 + 2 = 3 Marks]

Problem Statement (2083 Baishakh, Q2)

Differentiate between analog and digital signal. Convert $(159)_{10}$ to excess-3 code.

Part (a): Comparison between Analog and Digital Signals [1 Mark]

Parameter	Analog Signal	Digital Signal
Definition	Continuous in both time and amplitude over a continuous range.	Discrete in time and quantized into distinct binary logic levels ('0' and '1').
Information Representation	Expressed as continuous physical variations (voltage, current, pressure).	Expressed as discrete binary words / pulses.
Noise Immunity	Low; noise and distortion add directly to the signal and accumulate.	High; noise margins allow threshold regeneration via digital buffers without degradation.
Storage & Processing	Bulky magnetic tape, analog filters; prone to aging and drift.	Stored compactly in semiconductor memory / disk; easily processed by microprocessors and DSPs.
Error Correction	Extremely difficult to detect and correct transmission errors.	Error detection and correction codes (Parity, CRC, Hamming codes) easily implemented.

Part (b): Conversion of $(159)_{10}$ to Excess-3 Code [2 Marks] The **Excess-3 (XS-3)** code is an unweighted, self-complementing binary-coded decimal code. To convert a decimal number to Excess-3:

1. Add 3 (0011_2) to each decimal digit individually.
2. Convert each resulting sum (0 to 12) into its 4-bit binary representation.

For $(159)_{10}$:

- **Hundreds digit (1):** $1 + 3 = 4 \Rightarrow 0100_2$
- **Tens digit (5):** $5 + 3 = 8 \Rightarrow 1000_2$
- **Units digit (9):** $9 + 3 = 12 \Rightarrow 1100_2$

Final Verified Result

$$(159)_{10} = (0100 \ 1000 \ 1100)_{\text{Excess-3}}$$

1.0.3 Question 3: Universal Logic Gates Schematics [3 Marks]

Problem Statement (2083 Baishakh, Q3)

Show that NAND and NOR gates are universal gates.

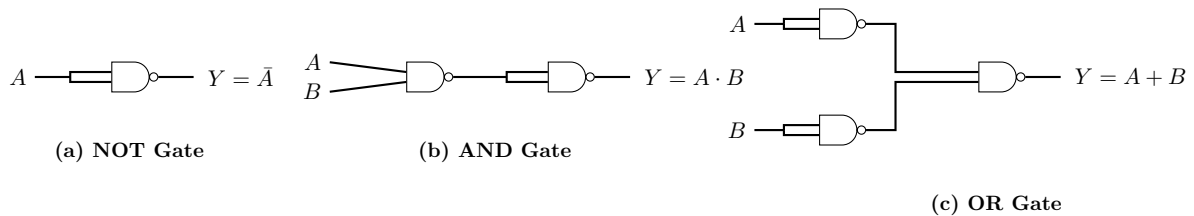
A logic gate is termed **universal** if any arbitrary Boolean function (or the complete set of primitive logic operations {NOT, AND, OR}) can be synthesized solely using that gate type without requiring any additional gates.

1. Realization of Basic Gates Using NAND Gates Only:

- **NOT Gate:** Tie both inputs together: $Y = \overline{A \cdot A} = \bar{A}$.

- **AND Gate:** A NAND gate followed by a NAND-inverter: $Y = \overline{\overline{A \cdot B}} = A \cdot B$.
- **OR Gate:** By De Morgan's Law, $A + B = \overline{\overline{A} \cdot \overline{B}}$. Invert A and B with NAND inverters, then apply to a NAND gate: $Y = \overline{\overline{A} \cdot \overline{B}} = A + B$.

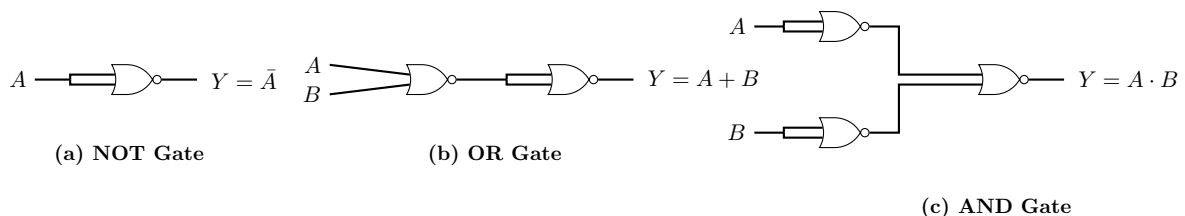
Logic Schematic / Circuit Diagram: Basic Logic Gates Synthesized from NAND Gates



2. Realization of Basic Gates Using NOR Gates Only:

- **NOT Gate:** Tie both inputs together: $Y = \overline{A + A} = \overline{A}$.
- **OR Gate:** A NOR gate followed by a NOR-inverter: $Y = \overline{\overline{A + B}} = A + B$.
- **AND Gate:** By De Morgan's Law, $A \cdot B = \overline{\overline{A} + \overline{B}}$. Invert A and B with NOR inverters, then apply to a NOR gate: $Y = \overline{\overline{A} + \overline{B}} = A \cdot B$.

Logic Schematic / Circuit Diagram: Basic Logic Gates Synthesized from NOR Gates



Final Verified Result

Since both NAND and NOR gates can independently synthesize the complete set of Boolean operators {NOT, AND, OR}, they are universally complete logic gates.

1.0.4 Question 4: 4-Variable K-Map Minimization & Circuit [5 Marks]

Problem Statement (2083 Baishakh, Q4)

Minimize the function $F(A, B, C, D) = \sum m(1, 2, 4, 5, 6, 8, 10, 11, 13, 15)$ using K-Map and realize it with suitable logic gates.

1. Karnaugh Map Plotting (4-Variable K-Map): The given minterms are plotted into the 4×4 Gray-coded map:

$AB \setminus CD$	00	01	11	10
00	0 (m_0)	1 (m_1)	0 (m_3)	1 (m_2)
01	1 (m_4)	1 (m_5)	0 (m_7)	1 (m_6)
11	0 (m_{12})	1 (m_{13})	1 (m_{15})	0 (m_{14})
10	1 (m_8)	0 (m_9)	1 (m_{11})	1 (m_{10})

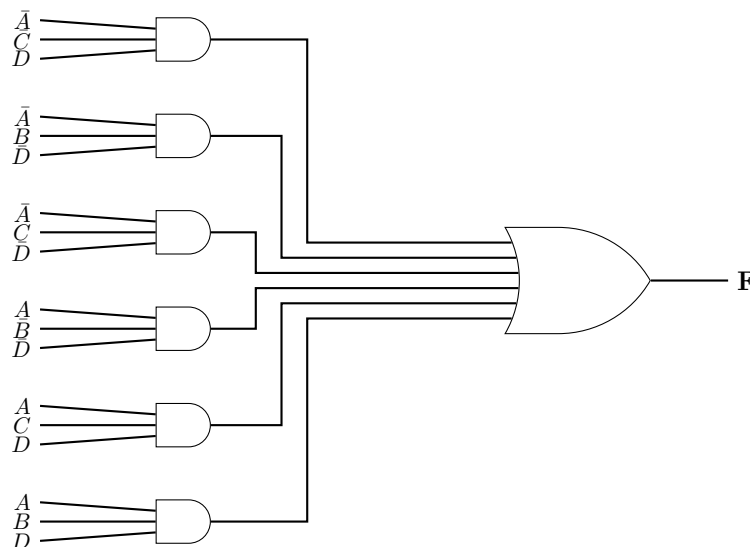
2. Systematic Grouping (Prime Implicants):

- Group 1 (Quad in column 01):** Minterms (m_1, m_5, m_{13}, m_{15}) $\Rightarrow \bar{C}D$.
- Group 2 (Pair in row 01):** Minterms (m_4, m_6) $\Rightarrow \bar{A}B\bar{D}$.
- Group 3 (Pair in column 10):** Minterms (m_2, m_6) $\Rightarrow \bar{A}C\bar{D}$.
- Group 4 (Pair in row 10):** Minterms (m_8, m_{10}) $\Rightarrow A\bar{B}\bar{D}$.
- Group 5 (Pair in column 11):** Minterms (m_{11}, m_{15}) $\Rightarrow ACD$.
- Group 6 (Pair in row 11):** Minterms (m_{13}, m_{15}) $\Rightarrow ABD$.

The minimal Sum of Products (SOP) expression is:

$$F(A, B, C, D) = \bar{A}\bar{C}D + \bar{A}B\bar{D} + \bar{A}C\bar{D} + A\bar{B}\bar{D} + ACD + ABD$$

Logic Schematic / Circuit Diagram: Two-Level AND-OR Logic Realization for Minimized $F(A, B, C, D)$



Final Verified Result

$$F(A, B, C, D) = \bar{A}\bar{C}D + \bar{A}B\bar{D} + \bar{A}C\bar{D} + A\bar{B}\bar{D} + ACD + ABD$$

1.0.5 Question 5: Octal-to-Binary Encoder Logic Circuit [2 + 4 = 6 Marks]

Problem Statement (2083 Baishakh, Q5)

Differentiate between encoder and decoder. Design an octal-to-binary encoder with necessary diagrams.

Part (a): Comparison between Encoder and Decoder [2 Marks]

Feature	Encoder	Decoder
Functional Operation	Converts 2^n (or fewer) input active lines into an n -bit coded binary output.	Converts an n -bit coded binary input into 2^n unique active output lines.
Inputs / Outputs	2^n input lines, n output lines.	n input lines, 2^n output lines.
Internal Logic Gate	Constructed using OR gates.	Constructed using AND / NAND gates.
Typical Applications	Keyboards, priority encoders, Flash Analog-to-Digital Converters (ADCs).	Memory address decoding, 7-segment display drivers, demultiplexers.

Part (b): Design of Octal-to-Binary Encoder (8-to-3 Encoder) [4 Marks] An octal-to-binary encoder accepts 8 active-high input lines $D_0, D_1, \dots, D_7$ (where exactly one input is HIGH at any instant) and produces a 3-bit binary equivalent A, B, C (A is MSB, C is LSB).

Truth Table:

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A (MSB)	B	C (LSB)
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Boolean Output Equations: Output bits are HIGH whenever any input corresponding to that binary position is active:

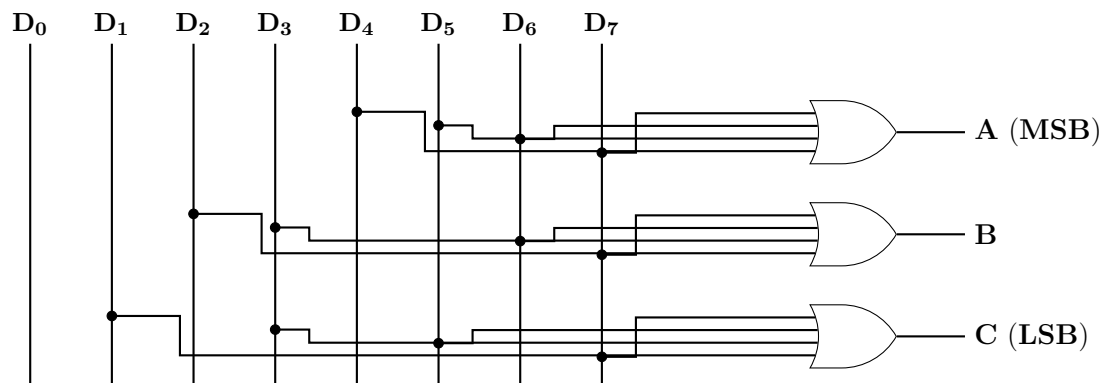
$$A = D_4 + D_5 + D_6 + D_7$$

$$B = D_2 + D_3 + D_6 + D_7$$

$$C = D_1 + D_3 + D_5 + D_7$$

(Note: D_0 is not required in the equations since $A = B = C = 0$ when $D_0 = 1$.)

Logic Schematic / Circuit Diagram: Octal-to-Binary (8-to-3) Encoder Logic Schematic



Final Verified Result

$$A = D_4 + D_5 + D_6 + D_7, \quad B = D_2 + D_3 + D_6 + D_7, \quad C = D_1 + D_3 + D_5 + D_7$$

1.0.6 Question 6: Full Subtractor Using Multiplexers [4 Marks]

Problem Statement (2083 Baishakh, Q6)

Implement a full subtractor circuit using multiplexer.

A full subtractor performs subtraction on three 1-bit inputs: Minuend (A), Subtrahend (B), and Borrow In (B_{in}). It generates Difference (D) and Borrow Out (B_{out}).

Truth Table:

A	B	B_{in}	D (Difference)	B_{out} (Borrow)
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

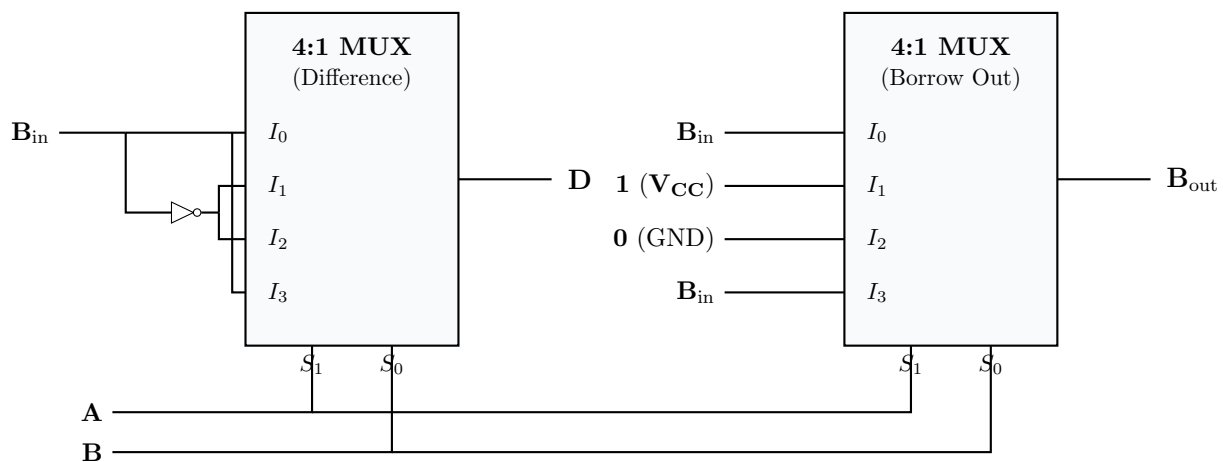
Minterm canonical forms:

$$D(A, B, B_{in}) = \sum m(1, 2, 4, 7), \quad B_{out}(A, B, B_{in}) = \sum m(1, 2, 3, 7)$$

Multiplexer Implementation using Two 4 : 1 MUXes: Let select lines be $S_1 = A$ and $S_0 = B$, and data inputs be expressed in terms of B_{in} :

Select Inputs (A, B)	Data Channel	Difference (D)	Borrow Out (B_{out})
00	I_0	B_{in}	B_{in}
01	I_1	$\bar{B}_{in}$	1 (V_{CC})
10	I_2	$\bar{B}_{in}$	0 (GND)
11	I_3	B_{in}	B_{in}

Logic Schematic / Circuit Diagram: Full Subtractor Implementation Using Two 4:1 Multiplexers



Final Verified Result

Difference (D): $I_0 = B_{in}, I_1 = \bar{B}_{in}, I_2 = \bar{B}_{in}, I_3 = B_{in}$.

Borrow Out (B_{out}): $I_0 = B_{in}, I_1 = 1, I_2 = 0, I_3 = B_{in}$, with select lines $S_1 = A, S_0 = B$.

1.0.7 Question 7: D Flip-Flop to JK Flip-Flop Conversion Schematic [2 + 5 = 7 Marks]

Problem Statement (2083 Baishakh, Q7)

Differentiate between latch and flipflop. Modify a D flip-flop such that it functions as a JK flip-flop.

Part (a): Comparison between Latch and Flip-Flop [2 Marks]

Parameter	Latch	Flip-Flop
Triggering Method	Level-triggered: transparent to input changes as long as Enable (EN) is active HIGH/LOW.	Edge-triggered: state transition occurs only during an active clock transition ($\uparrow$ positive or $\downarrow$ negative edge).
Clock Requirement	Does not require a periodic clock signal; operates on control/enable gates.	Requires a synchronous periodic clock signal for synchronized state updates.
Race Condition	Highly susceptible to race-around conditions when pulse width exceeds propagation delay.	Immune to race-around condition because input sampling occurs only during infinitesimal clock edges.
Circuit Role	Asynchronous registers, temporary storage buffers.	Synchronous counters, state machines, pipelines, CPU registers.

Part (b): Conversion of D Flip-Flop to JK Flip-Flop [5 Marks]

1. **Target Behavior:** JK Flip-Flop with inputs J, K , present state Q_n , and next state Q_{n+1} .
2. **Available Device:** D Flip-Flop whose characteristic equation is $Q_{n+1} = D$. Thus, $D = Q_{n+1}$.

Conversion Table:

J	K	Q_n	Q_{n+1}	D Input ($D = Q_{n+1}$)
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

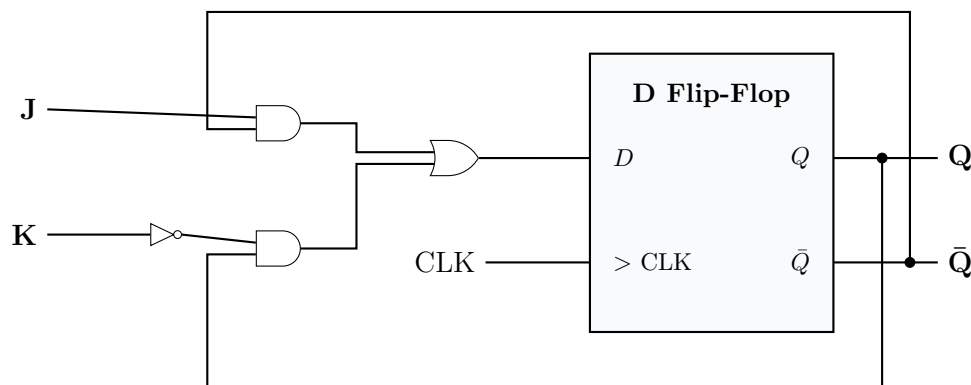
K-Map for Input $D(J, K, Q_n)$:

$JK \setminus Q_n$	0	1
00	0	1
01	0	0
11	1	0
10	1	1

- Group 1: Minterms for $Q_n = 0$ where $J = 1 \implies J\bar{Q}_n$
- Group 2: Minterms for $Q_n = 1$ where $K = 0 \implies \bar{K}Q_n$

$$D = J\bar{Q} + \bar{K}Q$$

Logic Schematic / Circuit Diagram: JK Flip-Flop Realized by Modifying a D Flip-Flop



Final Verified Result

$$D = J\bar{Q} + \bar{K}Q$$

A D flip-flop is transformed into a JK flip-flop by driving its D input with the combinational network $J\bar{Q} + \bar{K}Q$.

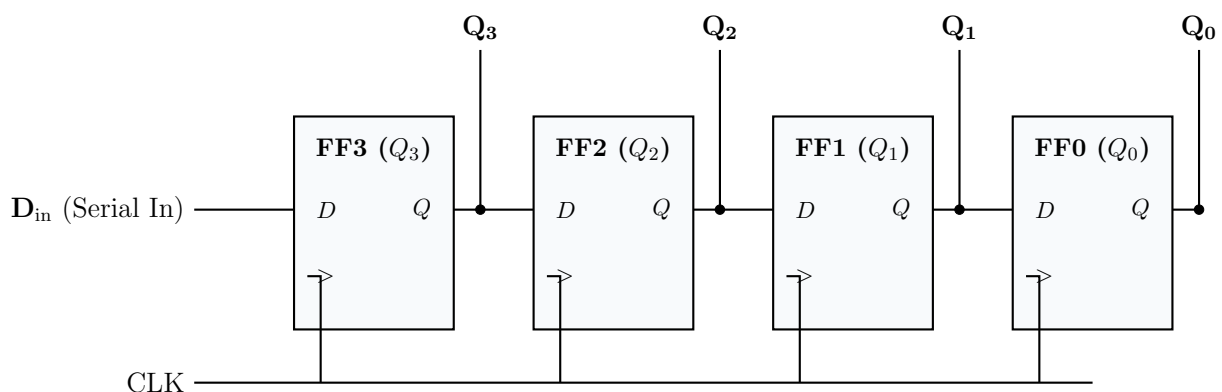
1.0.8 Question 8: 4-Bit SIPO Shift Register Schematic & Timing Diagram [5 Marks]

Problem Statement (2083 Baishakh, Q8)

Explain the working of 4-bit SIPO register with timing diagram of 1010 data input.

A **Serial-In Parallel-Out (SIPO)** shift register accepts binary data sequentially (bit-by-bit) over a single input line, shifting it through cascaded flip-flops on each clock edge. After 4 clock cycles, all 4 bits are simultaneously accessible on parallel output pins Q_3, Q_2, Q_1, Q_0 .

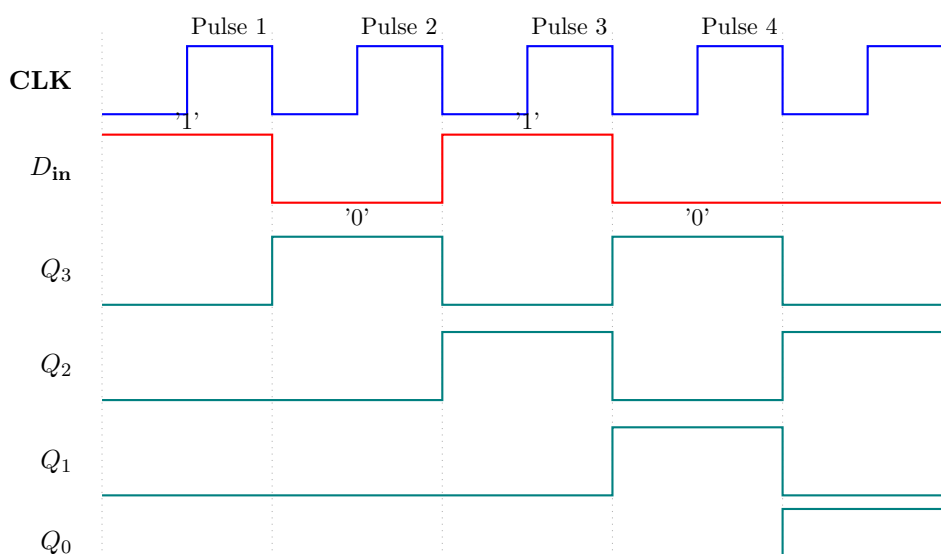
Logic Schematic / Circuit Diagram: 4-Bit Serial-In Parallel-Out (SIPO) Shift Register



Step-by-Step Operation for Input Sequence 1010_2 (MSB entered first): Assume initial contents are $Q_3Q_2Q_1Q_0 = 0000_2$.

Clock Pulse	Serial Input (D_{in})	Q_3	Q_2	Q_1	Q_0	Remarks
Initial	—	0	0	0	0	Cleared state
CLK 1	1 (MSB)	1	0	0	0	First bit enters FF3
CLK 2	0	0	1	0	0	Shift right, second bit enters
CLK 3	1	1	0	1	0	Shift right, third bit enters
CLK 4	0 (LSB)	0	1	0	1	Complete word loaded (1010_2)

Logic Schematic / Circuit Diagram: Timing Waveforms for 4-Bit SIPO Shift Register Loading 1010_2



Final Verified Result

After 4 clock pulses, serial word 1010_2 is converted to parallel outputs: $Q_3Q_2Q_1Q_0 = 0101_2$ (or 1010_2 when read in arrival sequence $Q_0Q_1Q_2Q_3$).

1.0.9 Question 9: Asynchronous BCD (Decade) Counter Schematic [5 Marks]

Problem Statement (2083 Baishakh, Q9)

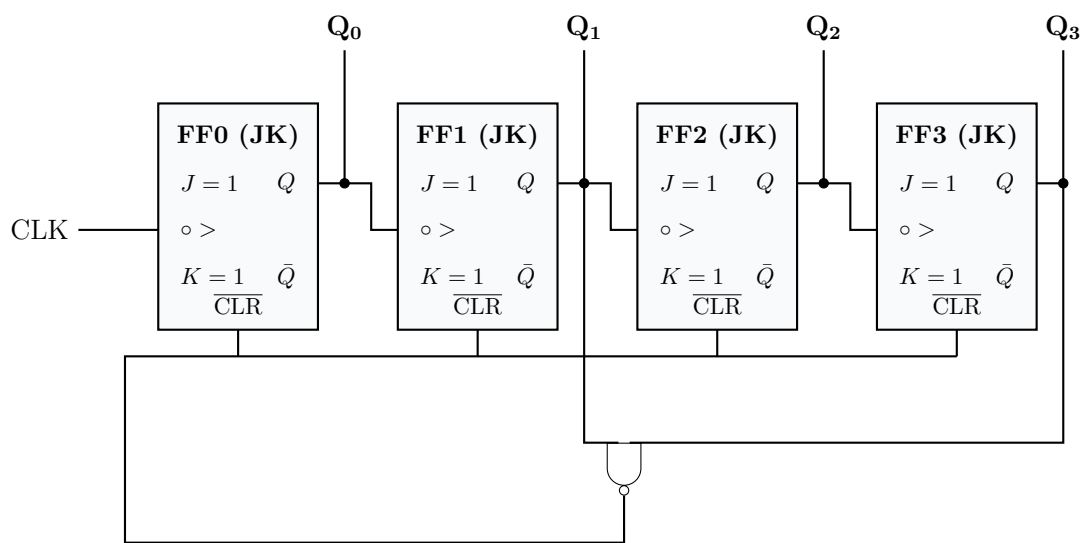
Describe the operation of asynchronous BCD (decade) counter with necessary diagrams.

An **Asynchronous BCD (Decade) Counter** (Mod-10 ripple counter) counts through 10 discrete binary states from 0000_2 (0_{10}) up to 1001_2 (9_{10}) and automatically recycles to 0000_2 on the tenth clock pulse.

Design Principle:

- 4 negative edge-triggered JK (or T) flip-flops are connected in toggle mode ($J = K = 1$).
- Ripple clocking: $Q_0 \rightarrow \text{CLK}_1$, $Q_1 \rightarrow \text{CLK}_2$, $Q_2 \rightarrow \text{CLK}_3$.
- Modulo reset: At the 10th clock pulse, the count momentarily enters transient state 1010_2 ($Q_3 = 1, Q_2 = 0, Q_1 = 1, Q_0 = 0$).
- A 2-input NAND gate detects $Q_3 \cdot Q_1 = 1$ and outputs an active-LOW reset pulse ($\overline{\text{CLR}} = 0$) to all flip-flops:

$$\overline{\text{CLR}} = \overline{Q_3 \cdot Q_1}$$

Logic Schematic / Circuit Diagram: Asynchronous BCD (Decade) Counter Logic Circuit**Final Verified Result**

The asynchronous BCD counter cycles through states $0000 \rightarrow 0001 \rightarrow \dots \rightarrow 1001 \rightarrow (1010 \text{ glitch}) \xrightarrow{\overline{\text{CLR}}=0} 0000$, implementing a stable Mod-10 frequency divider.

1.0.10 Question 10: Synchronous Mod-10 UP Counter Schematic [7 Marks]**Problem Statement (2083 Baishakh, Q10)**

Design the synchronous mod-10 up counter using T flip-flop and draw its timing diagram also.

1. State Transition and T Excitation Table ($T = Q_n \oplus Q_{n+1}$):

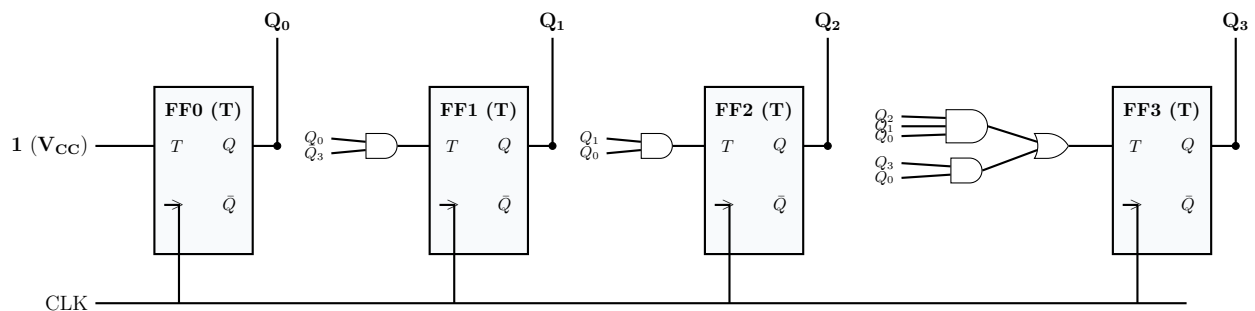
State	Present State				Next State				T Inputs			
	Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	0	1	0	0	0	1
1	0	0	0	1	0	0	1	0	0	0	1	1
2	0	0	1	0	0	0	1	1	0	0	0	1
3	0	0	1	1	0	1	0	0	0	1	1	1
4	0	1	0	0	0	1	0	1	0	0	0	1
5	0	1	0	1	0	1	1	0	0	0	1	1
6	0	1	1	0	0	1	1	1	0	0	0	1
7	0	1	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	1	0	0	1	0	0	0	1
9	1	0	0	1	0	0	0	0	1	0	0	1

(Unused states 10 to 15 are treated as Don't Cares X).

2. K-Map Derivations for T Inputs:

- $T_0 = 1$ (Q_0 toggles on every single clock pulse).
- $T_1 = \bar{Q}_3 Q_0$ (Q_1 toggles when $Q_0 = 1$ except when transitioning from 9 to 0).
- $T_2 = Q_1 Q_0$ (Q_2 toggles when $Q_1 Q_0 = 1$).
- $T_3 = Q_2 Q_1 Q_0 + Q_3 Q_0$ (Q_3 toggles at state 7 to become 1, and at state 9 to reset to 0).

Logic Schematic / Circuit Diagram: Synchronous Mod-10 UP Counter Logic Circuit



Final Verified Result

$$T_0 = 1, \quad T_1 = \bar{Q}_3 Q_0, \quad T_2 = Q_1 Q_0, \quad T_3 = Q_2 Q_1 Q_0 + Q_3 Q_0$$

1.0.11 Question 11: Synchronous FSM Sequence Detector '011' [7 Marks]

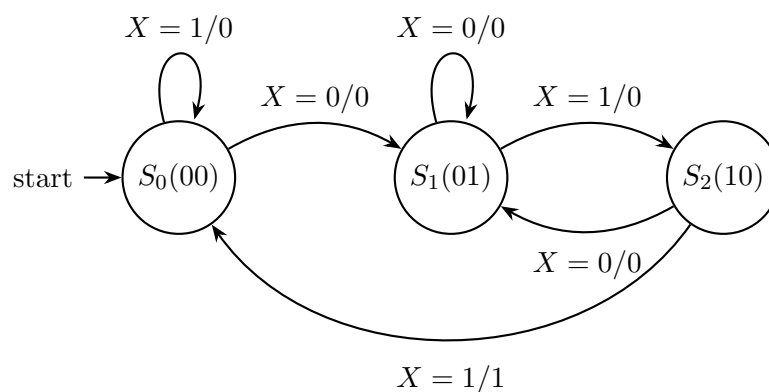
Problem Statement (2083 Baishakh, Q11)

Design a synchronous sequential machine that has 1-bit serial input X and output Z which will be high when the input contains the message '011' (Use T Flip-Flop).

1. State Definitions and State Diagram (Mealy Model):

- $S_0(00)$: Reset state / No relevant bit detected.
- $S_1(01)$: Sequence '0' detected.
- $S_2(10)$: Sequence '01' detected.
- When in S_2 and next bit $X = 1 \implies$ '011' detected $\implies Z = 1$, next state S_0 .

Logic Schematic / Circuit Diagram: State Transition Automaton for '011' Sequence Detector

2. State Transition and Excitation Table for T Flip-Flops (A, B):

Present State		Input X	Next State		Output Z	T Inputs	
A	B		A^+	B^+		T_A	T_B
0	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0
0	1	0	0	1	0	0	0
0	1	1	1	0	0	1	1
1	0	0	0	1	0	1	1
1	0	1	0	0	1	1	0
1	1	0	X	X	X	X	X
1	1	1	X	X	X	X	X

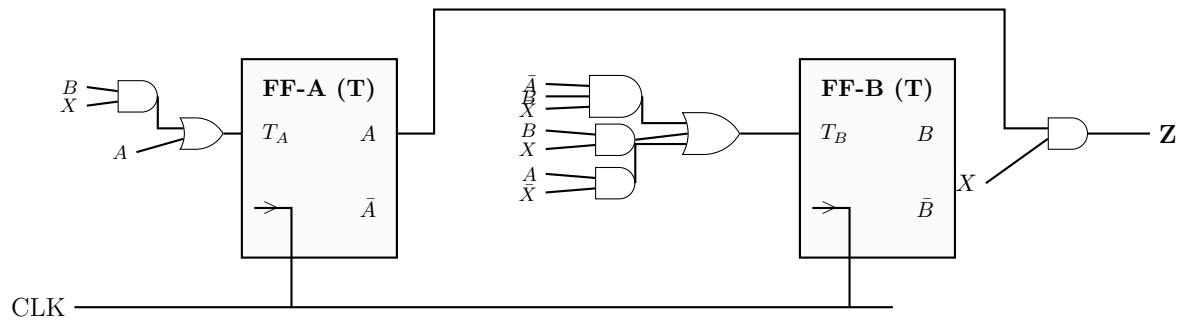
3. Minimized Equations via K-Maps:

$$T_A = A + BX$$

$$T_B = \bar{A}\bar{B}\bar{X} + BX + A\bar{X}$$

$$Z = AX$$

Logic Schematic / Circuit Diagram: Synchronous FSM Circuit Realization Using T Flip-Flops



Final Verified Result

$$T_A = A + BX, \quad T_B = \bar{A}\bar{B}\bar{X} + BX + A\bar{X}, \quad Z = AX$$

1.0.12 Question 12: Characteristics of Digital Logic Families [4 Marks]

Problem Statement (2083 Baishakh, Q12)

Explain about the main characteristics of digital logic families.

The performance and suitability of digital integrated circuit (IC) families are evaluated based on the following key operational parameters:

1. **Propagation Delay (t_{pd}):** The average time elapsed between the application of an input logic transition and the resulting output logic transition:

$$t_{pd} = \frac{t_{pLH} + t_{pHL}}{2}$$

where t_{pLH} is the low-to-high delay and t_{pHL} is the high-to-low delay. Lower propagation delay indicates higher operating speed.

2. **Power Dissipation (P_D):** The DC power consumed by the logic gate during static and dynamic operation, expressed in milliwatts (mW) or microwatts (μW):

$$P_D = V_{CC} \times I_{CC(\text{avg})}$$

3. **Speed-Power Product (Figure of Merit):** The product of propagation delay and power dissipation:

$$\text{SPP} = t_{pd} \times P_D \quad (\text{expressed in picojoules, pJ})$$

A lower SPP value signifies superior overall technological performance.

4. **Noise Margins (NM_H and NM_L):** The maximum allowable unwanted noise voltage superimposed on the input without causing an erroneous output change:

$$NM_H = V_{OH(\min)} - V_{IH(\min)}, \quad NM_L = V_{IL(\max)} - V_{OL(\max)}$$

5. **Fan-Out and Fan-In:**

- **Fan-Out:** Maximum number of standard logic gate inputs of the same family that an output can reliably drive without exceeding current limits.
- **Fan-In:** Number of independent input signals that a single gate can accommodate without degradation.

6. **Operating Temperature Range:** Standard commercial grade (0°C to 70°C) versus military grade (-55°C to $+125^\circ\text{C}$).

Comparative Summary of Major Digital Logic Families:

Parameter	Standard TTL (7400)	Schottky TTL (74S)	CMOS (74HC)	ECL (10K)
Propagation Delay (t_{pd})	10 ns	3 ns	8 ns	1 ns
Power Dissipation / Gate	10 mW	20 mW	0.001 mW (static)	25 mW
Speed-Power Product (SPP)	100 pJ	60 pJ	0.008 pJ	25 pJ
Noise Margin (NM)	0.4 V	0.3 V	> 1.0 V	0.25 V
Fan-Out	10	20	> 50	25

Final Verified Result

Digital logic families are comprehensively characterized by speed (t_{pd}), power consumption (P_D), noise margins (NM_H, NM_L), fan-out, and the Speed-Power Product figure of merit.

Chapter 2

2082 Bhadra Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)

Full Marks: 60

Programme: BEI, BCT

Pass Marks: 24

Year / Part: I / II

Time: 3 Hours

Subject: Digital Logic (ENEX 152 / EX 152)

Examination Type: Regular / Back

2.0.1 Question 1: Binary to Gray Code Conversion [2 + 2 = 4 Marks]

Problem Statement (2082 Bhadra, Q1)

Convert the following binary codes to gray codes:

(a) $(101101)_2$

(b) $(110001)_2$

Conversion Algorithm: Given an n -bit binary word $B = b_{n-1}b_{n-2}\dots b_1b_0$ and target Gray code $G = g_{n-1}g_{n-2}\dots g_1g_0$:

1. $g_{n-1} = b_{n-1}$ (The Most Significant Bit remains unchanged).
2. $g_i = b_{i+1} \oplus b_i$ for each subsequent bit position $i = n-2, n-3, \dots, 0$.

Part (a): Conversion of $(101101)_2$ Given $b_5b_4b_3b_2b_1b_0 = 101101_2$:

- $g_5 = b_5 = 1$
- $g_4 = b_5 \oplus b_4 = 1 \oplus 0 = 1$
- $g_3 = b_4 \oplus b_3 = 0 \oplus 1 = 1$

- $g_2 = b_3 \oplus b_2 = 1 \oplus 1 = 0$
- $g_1 = b_2 \oplus b_1 = 1 \oplus 0 = 1$
- $g_0 = b_1 \oplus b_0 = 0 \oplus 1 = 1$

Thus, $(101101)_2 = (111011)_{\text{Gray}}$.

Part (b): Conversion of $(110001)_2$ Given $b_5b_4b_3b_2b_1b_0 = 110001_2$:

- $g_5 = b_5 = 1$
- $g_4 = b_5 \oplus b_4 = 1 \oplus 1 = 0$
- $g_3 = b_4 \oplus b_3 = 1 \oplus 0 = 1$
- $g_2 = b_3 \oplus b_2 = 0 \oplus 0 = 0$
- $g_1 = b_2 \oplus b_1 = 0 \oplus 0 = 0$
- $g_0 = b_1 \oplus b_0 = 0 \oplus 1 = 1$

Thus, $(110001)_2 = (101001)_{\text{Gray}}$.

Final Verified Result

- (a) $(101101)_2 = (\mathbf{111011})_{\text{Gray}}$
 (b) $(110001)_2 = (\mathbf{101001})_{\text{Gray}}$

2.0.2 Question 2: 2's Complement & BCD Addition [1 + 2 = 3 Marks]

Problem Statement (2082 Bhadra, Q2)

What is 2's complement representation? Perform BCD addition of $(23)_{10} + (48)_{10}$.

Part (a): 2's Complement Representation [1 Mark] 2's complement is the standard system used in digital computers to represent signed integers. For an n -bit binary number X :

$$2\text{'s Complement of } X = 2^n - X = (1\text{'s Complement of } X) + 1$$

Key Advantages:

- Unambiguous representation of zero (single representation 0000_2 , unlike 1's complement which has $+0$ and -0).
- Simplifies hardware arithmetic: subtraction $A - B$ is performed as addition $A + (2\text{'s comp of } B)$ using standard binary adder circuits.

Part (b): BCD Addition of $23_{10} + 48_{10}$ [2 Marks] In BCD (8421 code), each decimal digit is encoded into 4 bits (0000_2 to 1001_2). If the sum of any decade exceeds 9 (1001_2) or produces a carry, a correction factor of +6 (0110_2) must be added.

- Represent numbers in BCD:

$$23_{10} = 0010\ 0011_{\text{BCD}}, \quad 48_{10} = 0100\ 1000_{\text{BCD}}$$

- **Step 1: Add Lower Decade (Units):**

$$\begin{array}{r} 0011 \quad (3_{10}) \\ + 1000 \quad (8_{10}) \\ \hline 1011_2 \quad (11_{10} > 9 \implies \text{Invalid BCD}) \end{array}$$

- **Step 2: Add Correction Factor (+0110₂):**

$$\begin{array}{r} 1011 \\ + 0110 \quad (+6_{10}) \\ \hline 0001_2 \quad \text{with Carry Out } C_{\text{out}} = 1 \text{ to Upper Decade} \end{array}$$

- **Step 3: Add Upper Decade (Tens) including Carry In:**

$$\begin{array}{r} 0010 \quad (2_{10}) \\ 0100 \quad (4_{10}) \\ + 0001 \quad (\text{Carry in} = 1) \\ \hline 0111_2 \quad (7_{10} \leq 9 \implies \text{Valid BCD}) \end{array}$$

Final Verified Result

$$(23)_{10} + (48)_{10} = 0111\ 0001_{\text{BCD}} = (71)_{10}$$

2.0.3 Question 3: De Morgan's Theorem with Diagrams [4 Marks]

Problem Statement (2082 Bhadra, Q3)

State and explain De Morgan's Theorem with truth table and necessary diagrams.

1. First Theorem (NAND-to-Inverted-OR Equivalence): The complement of a product of variables is equal to the sum of the complements of the variables:

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

2. Second Theorem (NOR-to-Inverted-AND Equivalence): The complement of a sum of variables is equal to the product of the complements of the variables:

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

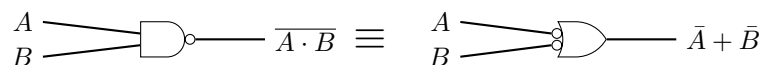
Truth Table Verification:

A	B	$\bar{A}$	$\bar{B}$	$\overline{A \cdot B}$	$\bar{A} + \bar{B}$	$\overline{A + B}$	$\bar{A} \cdot \bar{B}$
0	0	1	1	1	1	1	1
0	1	1	0	1	1	0	0
1	0	0	1	1	1	0	0
1	1	0	0	0	0	0	0

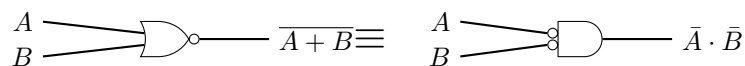
Columns 5 and 6 are identical, verifying Theorem 1. Columns 7 and 8 are identical, verifying Theorem 2.

Logic Schematic / Circuit Diagram: De Morgan's Equivalent Logic Gate Pairs

Theorem 1: $\overline{A \cdot B} = \bar{A} + \bar{B}$ (NAND $\equiv$ Inverted-Input OR)



Theorem 2: $\overline{A + B} = \bar{A} \cdot \bar{B}$ (NOR $\equiv$ Inverted-Input AND)



Final Verified Result

De Morgan's Laws establish dual relationships between AND and OR operations through inversion:
 $\overline{A \cdot B} = \bar{A} + \bar{B}$ and $\overline{A + B} = \bar{A} \cdot \bar{B}$.

2.0.4 Question 4: K-Map with Don't Cares & Circuit [5 Marks]

Problem Statement (2082 Bhadra, Q4)

Minimize the function $F(A, B, C, D) = \sum m(0, 1, 2, 4, 7, 8, 9, 10, 12, 15) + d(5, 11, 13)$ using K-Map and realize it with suitable logic gates.

1. K-Map Plotting (4-Variable K-Map):

$AB \setminus CD$	00	01	11	10
00	1 (m_0)	1 (m_1)	0 (m_3)	1 (m_2)
01	1 (m_4)	X (d_5)	1 (m_7)	0 (m_6)
11	1 (m_{12})	X (d_{13})	1 (m_{15})	0 (m_{14})
10	1 (m_8)	1 (m_9)	X (d_{11})	1 (m_{10})

2. Grouping Analysis:

1. Octet 1 (Columns $CD = 00$ and $CD = 01$):

Cells ($m_0, m_1, m_4, d_5, m_{12}, d_{13}, m_8, m_9$) $\Rightarrow \bar{C}$.

2. Quad 2 (Center and Column 11):

Cells (d_5, m_7, d_{13}, m_{15}) $\Rightarrow BD$.

3. Quad 3 (Four Corners):

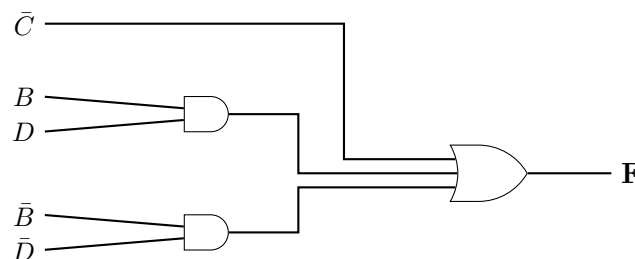
Cells (m_0, m_2, m_8, m_{10}) $\Rightarrow \bar{B}\bar{D}$.

Minimal Sum of Products (SOP):

$$F(A, B, C, D) = \bar{C} + BD + \bar{B}\bar{D}$$

(Note that $BD + \bar{B}\bar{D} = \overline{B \oplus D} = B \odot D$, the XNOR equivalence).

Logic Schematic / Circuit Diagram: Logic Circuit Realization for Minimized $F(A, B, C, D) = \bar{C} + BD + \bar{B}\bar{D}$



Final Verified Result

$$F(A, B, C, D) = \bar{C} + BD + \bar{B}\bar{D} = \bar{C} + (B \odot D)$$

2.0.5 Question 5: 3-Bit Magnitude Comparator Schematic [1 + 5 = 6 Marks]

Problem Statement (2082 Bhadra, Q5)

What is a magnitude comparator? Design a 3-bit magnitude comparator.

Part (a): Definition of Magnitude Comparator [1 Mark] A **magnitude comparator** is a combinational logic circuit that compares two binary numbers A and B , generating three mutually exclusive digital outputs indicating whether $A > B$, $A = B$, or $A < B$.

Part (b): Design of 3-Bit Magnitude Comparator [5 Marks] Let the two 3-bit input numbers be:

$$A = (A_2A_1A_0)_2, \quad B = (B_2B_1B_0)_2$$

where A_2, B_2 are the Most Significant Bits (MSBs) and A_0, B_0 are the Least Significant Bits (LSBs).

1. Equality Condition ($A = B$): Two numbers are equal if and only if all corresponding bits are identical:

$$x_i = A_i \odot B_i = A_i B_i + \bar{A}_i \bar{B}_i \quad \text{for } i = 0, 1, 2$$

Therefore, the equality output ($A = B$) is:

$$(A = B) = x_2 \cdot x_1 \cdot x_0$$

2. Greater Than Condition ($A > B$): A is strictly greater than B if:

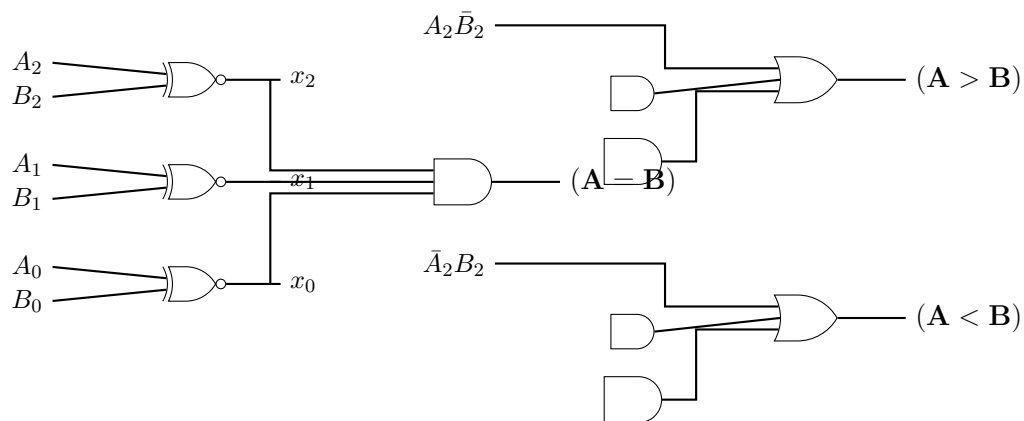
- The MSB of A is 1 and MSB of B is 0 ($A_2 > B_2 \implies A_2 \bar{B}_2$), OR
- MSBs are equal ($x_2 = 1$) and $A_1 > B_1$ ($A_1 \bar{B}_1$), OR
- MSBs and middle bits are equal ($x_2 x_1 = 1$) and $A_0 > B_0$ ($A_0 \bar{B}_0$).

$$(A > B) = A_2 \bar{B}_2 + x_2 A_1 \bar{B}_1 + x_2 x_1 A_0 \bar{B}_0$$

3. Less Than Condition ($A < B$): By symmetric logic:

$$(A < B) = \bar{A}_2 B_2 + x_2 \bar{A}_1 B_1 + x_2 x_1 \bar{A}_0 B_0$$

Logic Schematic / Circuit Diagram: 3-Bit Magnitude Comparator Logic Schematic



Final Verified Result

$$(A = B) = x_2 x_1 x_0$$

$$(A > B) = A_2 \bar{B}_2 + x_2 A_1 \bar{B}_1 + x_2 x_1 A_0 \bar{B}_0$$

$$(A < B) = \bar{A}_2 B_2 + x_2 \bar{A}_1 B_1 + x_2 x_1 \bar{A}_0 B_0$$

2.0.6 Question 6: Full Adder Using Multiplexers [4 Marks]

Problem Statement (2082 Bhadra, Q6)

Implement a full adder circuit using Multiplexer.

A full adder adds three 1-bit binary inputs: Augend (A), Addend (B), and Carry In (C_{in}). It generates Sum (S) and Carry Out (C_{out}).

Truth Table:

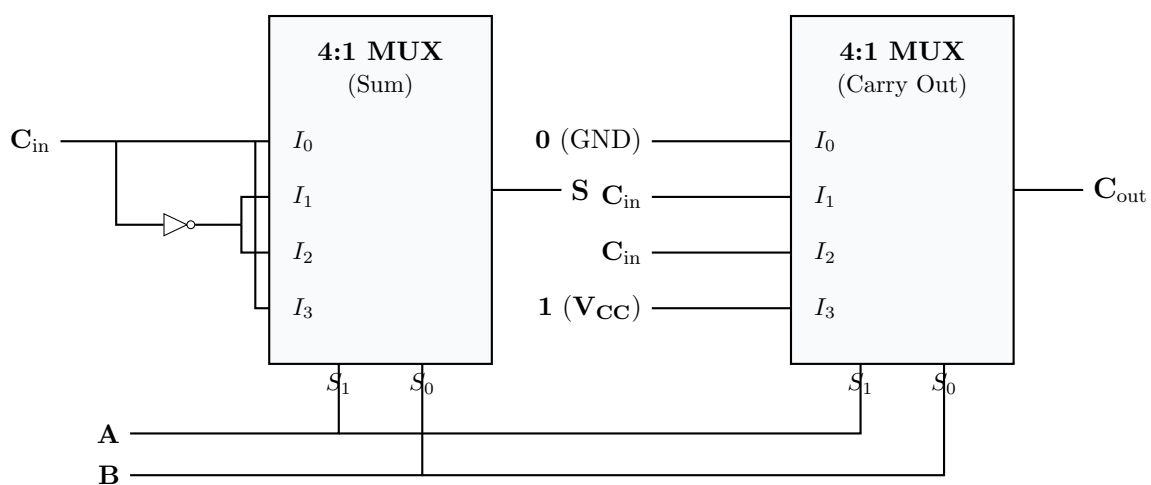
A	B	C_{in}	S (Sum)	C_{out} (Carry)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$S(A, B, C_{\text{in}}) = \sum m(1, 2, 4, 7), \quad C_{\text{out}}(A, B, C_{\text{in}}) = \sum m(3, 5, 6, 7)$$

Multiplexer Implementation using Two 4 : 1 MUXes: Let select lines be $S_1 = A, S_0 = B$:

Select Inputs (A, B)	Data Channel	Sum (S)	Carry Out (C_{out})
00	I_0	C_{in}	0 (GND)
01	I_1	$\bar{C}_{\text{in}}$	C_{in}
10	I_2	$\bar{C}_{\text{in}}$	C_{in}
11	I_3	C_{in}	1 (V_{CC})

Logic Schematic / Circuit Diagram: Full Adder Implementation Using Two 4:1 Multiplexers



Final Verified Result

Sum (S): $I_0 = C_{\text{in}}, I_1 = \bar{C}_{\text{in}}, I_2 = \bar{C}_{\text{in}}, I_3 = C_{\text{in}}$.

Carry Out (C_{out}): $I_0 = 0, I_1 = C_{\text{in}}, I_2 = C_{\text{in}}, I_3 = 1$, with select lines $S_1 = A, S_0 = B$.

2.0.7 Question 7: SR to JK Flip-Flop Conversion Schematic [2 + 5 = 7 Marks]

Problem Statement (2082 Bhadra, Q7)

Differentiate between combinational circuit and sequential circuit. With necessary steps and implementation, convert an SR flip-flop into a JK flip-flop.

Part (a): Combinational vs. Sequential Circuits [2 Marks]

Parameter	Combinational Circuit	Sequential Circuit
Output Dependence	Depends strictly on the present inputs at that instant.	Depends on both present inputs and the past history (internal stored state).
Memory Elements	Contains no memory elements or feedback paths.	Contains memory elements (flip-flops, latches) with feedback loops.
Clock Synchronization	Clock signal is not required; operates asynchronously.	Usually driven by a synchronous master clock signal.
Design Complexity	Simple (Boolean algebraic minimization, K-maps).	Complex (State tables, state diagrams, FSM models).
Examples	Adders, Subtractors, Multiplexers, Decoders.	Flip-flops, Counters, Shift registers, CPUs.

Part (b): Conversion of SR Flip-Flop into JK Flip-Flop [5 Marks]

1. **Target:** JK Flip-Flop with inputs J, K , present state Q_n , next state Q_{n+1} .
2. **Available:** SR Flip-Flop with excitation requirements:

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

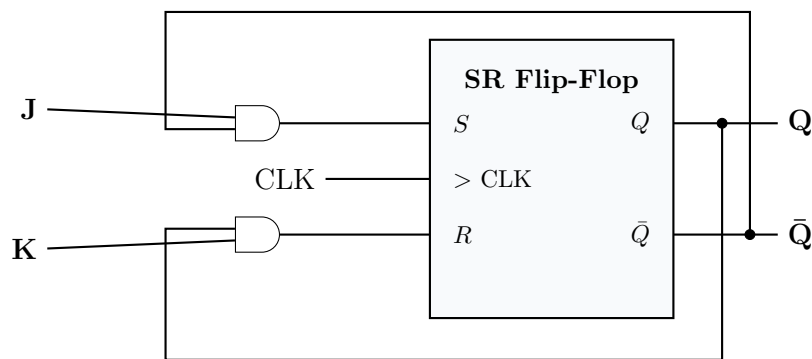
Conversion Table:

J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

K-Map Minimization for S and R :

- For $S(J, K, Q_n)$: Minterms are m_4, m_6 ; Don't cares are $d_1, d_5 \implies \mathbf{S = J\bar{Q}_n}$.
- For $R(J, K, Q_n)$: Minterms are m_3, m_7 ; Don't cares are $d_0, d_2 \implies \mathbf{R = KQ_n}$.

Logic Schematic / Circuit Diagram: SR Flip-Flop Converted to JK Flip-Flop



Final Verified Result

$$S = J\bar{Q}, \quad R = KQ$$

An SR flip-flop is converted into a JK flip-flop by connecting $S = J\bar{Q}$ and $R = KQ$. This eliminates the invalid state ($S = R = 1$) and enables toggling ($J = K = 1$).

2.0.8 Question 8: Types of Shift Registers & 4-Bit Johnson Counter [2 + 3 = 5 Marks]

Problem Statement (2082 Bhadra, Q8)

Write briefly about different types of shift registers. With necessary circuit and timing diagrams, explain the operation of how the shift register is used as a Johnson's Counter.

Part (a): Classification of Shift Registers [2 Marks]

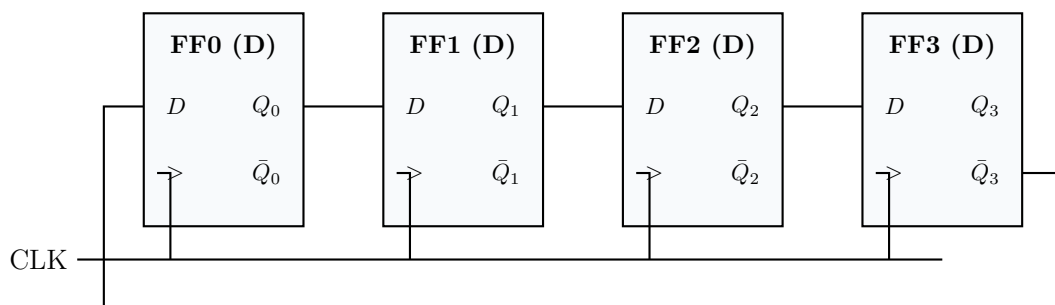
1. **SISO (Serial-In Serial-Out):** Data loaded and retrieved serially. Used for time delay generation.
2. **SIPO (Serial-In Parallel-Out):** Data loaded serially and read out in parallel. Used for serial-to-parallel data conversion.
3. **PISO (Parallel-In Serial-Out):** Data loaded simultaneously in parallel and shifted out serially. Used in serial transmitters (e.g. UART).
4. **PIPO (Parallel-In Parallel-Out):** Data loaded and read out simultaneously. Used as general-purpose internal CPU registers.
5. **Bidirectional / Universal Shift Register:** Can shift data left, right, load parallel data, or hold state based on mode select lines.

Part (b): 4-Bit Johnson (Twisted Ring) Counter [3 Marks] A Johnson Counter (or Moebius Counter) is constructed from an n -bit shift register by connecting the inverted output of the last stage ($\bar{Q}_3$) back to the data input of the first stage (D_0):

$$D_0 = \bar{Q}_3$$

An n -bit Johnson counter yields $2n$ unique states ($2 \times 4 = 8$ states for a 4-bit register), compared to n states for a standard ring counter.

Logic Schematic / Circuit Diagram: 4-Bit Johnson (Twisted-Ring) Counter Circuit Schematic



State Sequence Table (8 Distinct States):

Clock Pulse	Q_0	Q_1	Q_2	Q_3	Next Input $D_0 = \bar{Q}_3$
0 (Initial)	0	0	0	0	$D_0 = \bar{0} = 1$
1	1	0	0	0	$D_0 = \bar{0} = 1$
2	1	1	0	0	$D_0 = \bar{0} = 1$
3	1	1	1	0	$D_0 = \bar{0} = 1$
4	1	1	1	1	$D_0 = \bar{1} = 0$
5	0	1	1	1	$D_0 = \bar{1} = 0$
6	0	0	1	1	$D_0 = \bar{1} = 0$
7	0	0	0	1	$D_0 = \bar{1} = 0$
8 (Recycles)	0	0	0	0	Recycles to state 0

Final Verified Result

A 4-bit Johnson counter provides $2n = 8$ decoding states with adjacent states differing by only a single bit (Gray-code property), operating as a divide-by-8 frequency scaler.

2.0.9 Question 9: 3-Bit Ripple UP Counter with Positive Edge Triggering [5 Marks]

Problem Statement (2082 Bhadra, Q9)

Describe the operation of 3-bit ripple up counter with positive edge triggered clock.

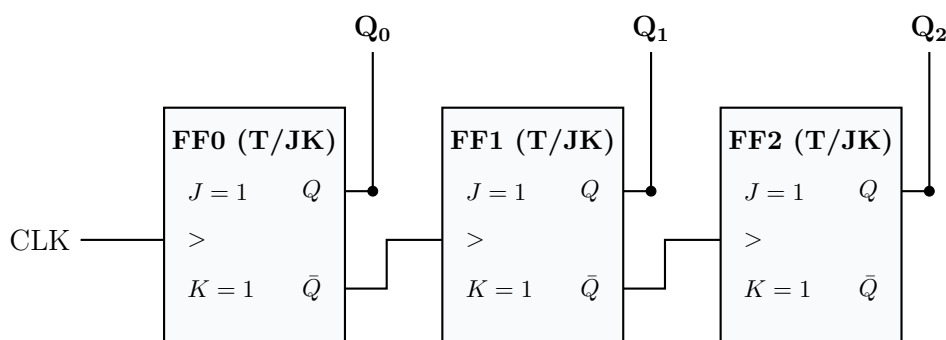
A **3-Bit Ripple UP Counter** counts through 8 binary states from 000_2 to 111_2 .

Clocking Principle for Positive-Edge Triggered Flip-Flops:

- In a ripple UP counter, the next higher bit Q_{i+1} must toggle whenever the current bit Q_i transitions from $1 \rightarrow 0$.
- Since the flip-flops are **positive edge-triggered** ($\uparrow$, triggering on $0 \rightarrow 1$), the transition $Q_i : 1 \rightarrow 0$ corresponds to inverted output $\bar{Q}_i : 0 \rightarrow 1$.
- Therefore, to achieve UP counting with positive-edge triggered flip-flops, the clock input of each subsequent flip-flop must be driven by the **complementary output** $\bar{Q}$ of the preceding stage:

$$\text{CLK}_{i+1} = \bar{Q}_i$$

Logic Schematic / Circuit Diagram: 3-Bit Positive-Edge Triggered Ripple UP Counter



Final Verified Result

By connecting $\text{CLK}_{i+1} = \bar{Q}_i$, the positive-edge triggered flip-flops toggle on every $1 \rightarrow 0$ transition of Q_i , producing the standard binary UP counting sequence: **000 $\rightarrow$ 001 $\rightarrow$ 010 $\rightarrow$ 011 $\rightarrow$ 100 $\rightarrow$ 101 $\rightarrow$ 110 $\rightarrow$ 111 $\rightarrow$ 000.**

2.0.10 Question 10: Synchronous Mod-6 Counter Using SR Flip-Flops [7 Marks]**Problem Statement (2082 Bhadra, Q10)**

Design the synchronous mod-6 counter using S-R flip-flop and draw its timing diagram also.

A **Synchronous Mod-6 Counter** counts through 6 states: $000 \rightarrow 001 \rightarrow 010 \rightarrow 011 \rightarrow 100 \rightarrow 101 \rightarrow 000$. Requires $N = 3$ flip-flops ($2^2 < 6 \leq 2^3$). Unused states: 110, 111 (Don't Cares).

1. State Transition and SR Excitation Table:

State	Present State			Next State			SR Flip-Flop Inputs					
	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	S_2	R_2	S_1	R_1	S_0	R_0
0	0	0	0	0	0	1	0	X	0	X	1	0
1	0	0	1	0	1	0	0	X	1	0	0	1
2	0	1	0	0	1	1	0	X	X	0	1	0
3	0	1	1	1	0	0	1	0	0	1	0	1
4	1	0	0	1	0	1	X	0	0	X	1	0
5	1	0	1	0	0	0	0	1	0	X	0	1

2. Minimized Equations via K-Maps:

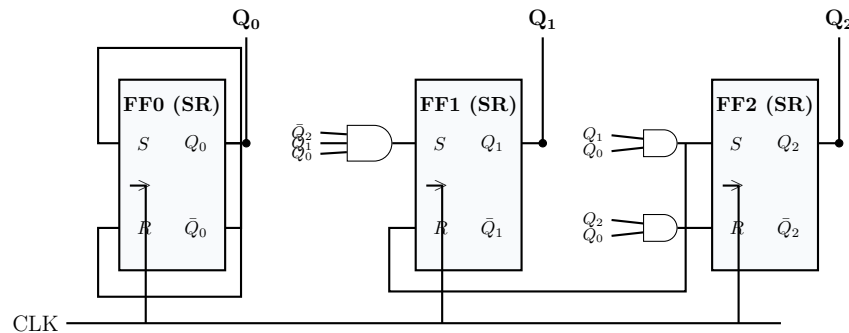
$$\begin{aligned}
 S_0 &= \bar{Q}_0, & R_0 &= Q_0 \\
 S_1 &= \bar{Q}_2 \bar{Q}_1 Q_0, & R_1 &= Q_1 Q_0 \\
 S_2 &= Q_1 Q_0, & R_2 &= Q_2 Q_0
 \end{aligned}$$

3. Self-Starting & Lock-out Verification:

- If in state 110_2 : $S_0 = 1, R_0 = 0 \Rightarrow Q_0^+ = 1$; $S_1 = 0, R_1 = 0 \Rightarrow Q_1^+ = 1$; $S_2 = 0, R_2 = 0 \Rightarrow Q_2^+ = 1 \Rightarrow$ Next State 111_2 .
- If in state 111_2 : $S_0 = 0, R_0 = 1 \Rightarrow Q_0^+ = 0$; $S_1 = 0, R_1 = 1 \Rightarrow Q_1^+ = 0$; $S_2 = 0, R_2 = 1 \Rightarrow Q_2^+ = 0 \Rightarrow$ Next State 000_2 (Valid state!).

Thus, the counter is completely self-starting and cannot get trapped in unused states.

Logic Schematic / Circuit Diagram: Synchronous Mod-6 Counter Logic Circuit Using SR Flip-Flops



Final Verified Result

$$S_0 = \bar{Q}_0, R_0 = Q_0, S_1 = \bar{Q}_2 \bar{Q}_1 Q_0, R_1 = Q_1 Q_0, S_2 = Q_1 Q_0, R_2 = Q_2 Q_0$$

2.0.11 Question 11: Synchronous FSM Sequence Detector '110' [7 Marks]

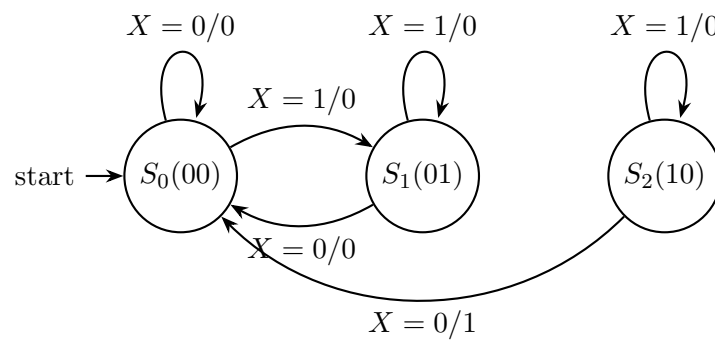
Problem Statement (2082 Bhadra, Q11)

Design a synchronous sequential machine that has 1-bit serial input X and output Z which will be high when the input contains the message '110' (Use SR Flip-Flop).

1. State Definitions (Mealy Model):

- $S_0(00)$: Reset state / No relevant bit detected.
- $S_1(01)$: Pattern '1' detected.
- $S_2(10)$: Pattern '11' detected.
- When in S_2 and $X = 0 \Rightarrow$ '110' detected $\Rightarrow Z = 1$, next state S_0 .

Logic Schematic / Circuit Diagram: State Diagram for '110' Sequence Detector



2. State Transition & SR Excitation Table (Flip-Flops A, B):

Present State		Input X	Next State		Output Z	SR Inputs			
A	B		A ⁺	B ⁺		S _A	R _A	S _B	R _B
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	0	X	1	0
0	1	0	0	0	0	0	X	0	1
0	1	1	1	0	0	1	0	0	1
1	0	0	0	0	1	0	1	0	X
1	0	1	1	0	0	X	0	0	X
1	1	0	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X

3. Minimized Equations via K-Maps:

$$\begin{aligned}
 S_A &= BX, & R_A &= \bar{X} \\
 S_B &= \bar{A}\bar{B}X, & R_B &= \bar{X} + B \\
 Z &= A\bar{X}
 \end{aligned}$$

Final Verified Result

$$S_A = BX, \quad R_A = \bar{X}, \quad S_B = \bar{A}\bar{B}X, \quad R_B = \bar{X} + B, \quad Z = A\bar{X}$$

2.0.12 Question 12: Frequency Counter Block Diagram [3 Marks]

Problem Statement (2082 Bhadra, Q12)

Write a short note on frequency counter.

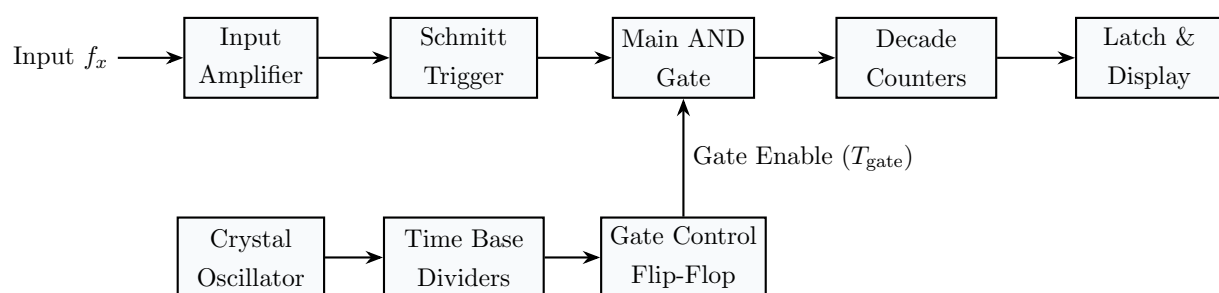
A **Digital Frequency Counter** is an electronic test instrument that accurately measures the frequency of an unknown periodic input signal by counting the number of cycles (N) that occur during a precisely controlled time window (T_{gate}):

$$f_x = \frac{N}{T_{\text{gate}}}$$

Core Functional Subsystems:

1. **Input Attenuator & Amplifier:** Scales input voltage to safe operational levels.
2. **Schmitt Trigger (Wave Shaper):** Converts arbitrary input waveforms (sinusoidal, triangular) into sharp, clean digital rectangular pulses.
3. **Crystal Oscillator & Time Base Generator:** Highly stable quartz crystal oscillator (e.g. 1 MHz or 10 MHz) paired with decade frequency dividers to produce accurate gate periods (1 ms, 10 ms, 0.1 s, 1 s).
4. **Main Gate (AND Gate):** Enabled for exactly T_{gate} , passing N input pulses to the counter.
5. **Decade Counter, Latch, & Display:** Counts incoming pulses, latches the result, and drives 7-segment digital displays.

Logic Schematic / Circuit Diagram: Functional Block Diagram of a Digital Frequency Counter



Final Verified Result

A frequency counter counts incoming signal transitions over a precision gate window T_{gate} , calculating frequency via $f = N/T_{\text{gate}}$.

Chapter 3

2082 Baishakh Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)	Full Marks: 60
Programme: BCT, BEI	Pass Marks: 24
Year / Part: I / II	Time: 3 Hours
Subject: Digital Logic (EX 152)	Examination Type: Back (New Course)

3.0.1 Question 1: ASCII Code & 2's Complement Addition [1 + 3 = 4 Marks]

Problem Statement (2082 Baishakh, Q1)

Define an ASCII code. Use 2's complement method to perform the following addition $(-28 + 12)_{10}$ in 8-bit signed number representation.

Part (a): Definition of ASCII Code [1 Mark] **ASCII** (*American Standard Code for Information Interchange*) is a standardized alphanumeric character encoding scheme:

- Standard ASCII uses a **7-bit binary code** to represent $2^7 = 128$ distinct characters, including English alphabets ($A - Z, a - z$), numerals ($0 - 9$), punctuation symbols, and non-printable control characters (e.g., CR, LF, ESC).
- Extended ASCII uses an **8-bit code** ($2^8 = 256$ characters) to accommodate graphical symbols and international characters.

Part (b): 2's Complement Addition of $(-28 + 12)_{10}$ in 8 Bits [3 Marks]

1. Representation of $+12_{10}$ in 8-bit binary:

$$+12_{10} = 0000\ 1100_2$$

2. Representation of -28_{10} in 8-bit 2's complement form:

$$+28_{10} = 0001\ 1100_2$$

Take 1's complement: 1's comp(0001 1100) = 1110 0011₂.

Add 1:

$$-28_{10} = 1110\ 0011 + 1 = \mathbf{1110\ 0100}_2$$

3. Perform 8-bit Binary Addition:

$$\begin{array}{r} 1110\ 0100 \quad (-28_{10}) \\ + 0000\ 1100 \quad (+12_{10}) \\ \hline \mathbf{1111\ 0000}_2 \quad (\text{Result in 2's complement form}) \end{array}$$

4. Verification and Decimal Interpretation:

- MSB = 1 $\implies$ Result is **negative**.
- Magnitude = 2's complement of $1111\ 0000_2 = (0000\ 1111)_2 + 1 = 0001\ 0000_2 = \mathbf{16}_{10}$.
- Therefore, result is $-\mathbf{16}_{10}$, which matches $-28 + 12 = -16$.

Final Verified Result

$$(-28 + 12)_{10} = (\mathbf{1111\ 0000})_2 = -16_{10}$$

3.0.2 Question 2: Boolean Algebra Proofs [2 + 2 = 4 Marks]

Problem Statement (2082 Baishakh, Q2)

Prove the following Boolean identities algebraically:

- (a) $AB + \bar{A}C + BC = AB + \bar{A}C$
- (b) $AB + C(A \oplus B) = BC + A(B + C)$

Part (a): Consensus Theorem Proof [2 Marks] To prove: $AB + \bar{A}C + BC = AB + \bar{A}C$.

$$\begin{aligned}
 \text{LHS} &= AB + \bar{A}C + BC \\
 &= AB + \bar{A}C + BC(1) && \text{(Identity Law: } X \cdot 1 = X) \\
 &= AB + \bar{A}C + BC(A + \bar{A}) && \text{(Complementarity Law: } A + \bar{A} = 1) \\
 &= AB + \bar{A}C + ABC + \bar{A}BC && \text{(Distributive Law)} \\
 &= (AB + ABC) + (\bar{A}C + \bar{A}BC) && \text{(Associative \& Commutative Laws)} \\
 &= AB(1 + C) + \bar{A}C(1 + B) && \text{(Distributive Law)} \\
 &= AB(1) + \bar{A}C(1) && \text{(Dominance Law: } 1 + X = 1) \\
 &= AB + \bar{A}C = \text{RHS} \quad \blacksquare
 \end{aligned}$$

Part (b): Identity Proof [2 Marks] To prove: $AB + C(A \oplus B) = BC + A(B + C)$.

$$\begin{aligned}
 \text{LHS} &= AB + C(A \oplus B) \\
 &= AB + C(A\bar{B} + \bar{A}B) && \text{(Definition of XOR: } A \oplus B = A\bar{B} + \bar{A}B) \\
 &= AB + A\bar{B}C + \bar{A}BC \\
 &= AB(1 + C) + A\bar{B}C + \bar{A}BC && \text{(Since } AB = AB(1 + C) = AB + ABC) \\
 &= AB + ABC + A\bar{B}C + \bar{A}BC \\
 &= AB + AC(B + \bar{B}) + BC(A + \bar{A}) \\
 &= AB + AC(1) + BC(1) \\
 &= AB + AC + BC
 \end{aligned}$$

Now expanding the Right Hand Side (RHS):

$$\begin{aligned}
 \text{RHS} &= BC + A(B + C) \\
 &= BC + AB + AC \\
 &= AB + AC + BC
 \end{aligned}$$

Since $\text{LHS} = \text{RHS} = AB + AC + BC$, the identity is proved. $\blacksquare$

Final Verified Result

Both Boolean identities are formally verified using fundamental axioms and laws of Boolean algebra.

3.0.3 Question 3: Octal Priority Encoder Schematic [2 + 6 = 8 Marks]

Problem Statement (2082 Baishakh, Q3)

What is a decoder? Design an octal priority encoder with neat circuit diagram.

Part (a): Definition of Decoder [2 Marks] A **decoder** is a combinational logic circuit with n input lines and 2^n unique output lines. It decodes an n -bit binary code word by asserting exactly one output line HIGH (or active LOW) corresponding to the decimal equivalent of the input binary code.

Part (b): Design of Octal Priority Encoder (8-to-3) [6 Marks] An **Octal Priority Encoder** accepts 8 input lines $D_0, D_1, \dots, D_7$ where input D_7 has the highest priority and D_0 has the lowest priority. If multiple inputs are HIGH simultaneously, the output binary code $A_2 A_1 A_0$ reflects the highest-indexed active input. A **Valid Output** bit (V) indicates whether any input is active.

Truth Table:

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A_2	A_1	A_0	V (Valid)
0	0	0	0	0	0	0	0	X	X	X	0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	0	1	X	0	0	1	1
0	0	0	0	0	1	X	X	0	1	0	1
0	0	0	0	1	X	X	X	0	1	1	1
0	0	0	1	X	X	X	X	1	0	0	1
0	0	1	X	X	X	X	X	1	0	1	1
0	1	X	X	X	X	X	X	1	1	0	1
1	X	X	X	X	X	X	X	1	1	1	1

Minimized Boolean Output Equations:

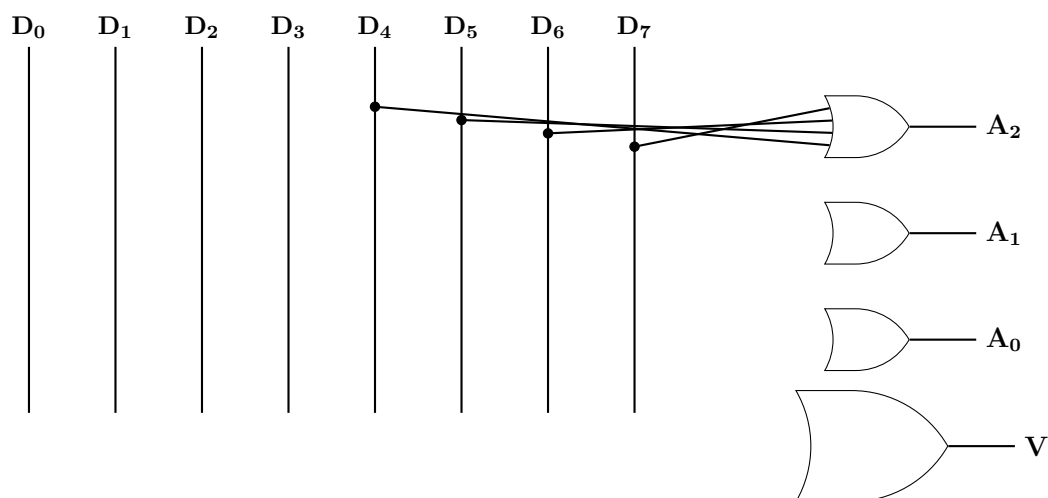
$$A_2 = D_4 + D_5 + D_6 + D_7$$

$$A_1 = D_6 + D_7 + \bar{D}_6 \bar{D}_5 (D_2 + D_3) = D_6 + D_7 + D_2 \bar{D}_4 \bar{D}_5 + D_3 \bar{D}_4 \bar{D}_5$$

$$A_0 = D_7 + D_5 \bar{D}_6 + D_3 \bar{D}_4 \bar{D}_6 + D_1 \bar{D}_2 \bar{D}_4 \bar{D}_6$$

$$V = D_0 + D_1 + D_2 + D_3 + D_4 + D_5 + D_6 + D_7$$

Logic Schematic / Circuit Diagram: Octal (8-to-3) Priority Encoder Circuit Schematic



Final Verified Result

$$A_2 = D_4 + D_5 + D_6 + D_7, \quad A_1 = D_6 + D_7 + \bar{D}_4 \bar{D}_5 (D_2 + D_3), \quad V = \sum_{i=0}^7 D_i$$

3.0.4 Question 4: 3x8 Decoder Function Realization & K-Map [3 + 3 = 6 Marks]

Problem Statement (2082 Baishakh, Q4)

Realize the Boolean function $X(A, B, C, D) = \sum m(0, 2, 3, 7, 8, 10, 11, 14, 15)$ using a single 3×8 decoder. Also simplify the logic function implementing K-map method.

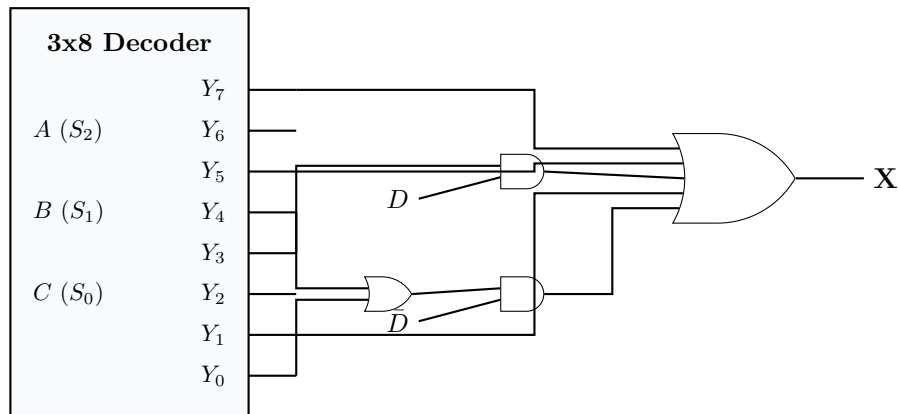
Part (a): Realization Using a Single 3×8 Decoder [3 Marks] Let variables A, B, C be connected to decoder select inputs S_2, S_1, S_0 , and let variable D serve as external data input:

- $Y_0 = \bar{A}\bar{B}\bar{C}$: includes $m_0(\bar{D}), m_1(D) \Rightarrow$ function has $m_0 \Rightarrow Y_0 \bar{D}$.
- $Y_1 = \bar{A}\bar{B}C$: includes $m_2(\bar{D}), m_3(D) \Rightarrow$ function has both $m_2, m_3 \Rightarrow Y_1$.
- $Y_2 = \bar{A}B\bar{C}$: includes $m_4, m_5 \Rightarrow 0$.
- $Y_3 = \bar{A}BC$: includes $m_6(\bar{D}), m_7(D) \Rightarrow$ function has $m_7 \Rightarrow Y_3 D$.
- $Y_4 = A\bar{B}\bar{C}$: includes $m_8(\bar{D}), m_9(D) \Rightarrow$ function has $m_8 \Rightarrow Y_4 \bar{D}$.
- $Y_5 = A\bar{B}C$: includes $m_{10}(\bar{D}), m_{11}(D) \Rightarrow$ function has both $m_{10}, m_{11} \Rightarrow Y_5$.
- $Y_6 = AB\bar{C}$: includes $m_{12}, m_{13} \Rightarrow 0$.
- $Y_7 = ABC$: includes $m_{14}(\bar{D}), m_{15}(D) \Rightarrow$ function has both $m_{14}, m_{15} \Rightarrow Y_7$.

Combining terms:

$$X = (Y_0 + Y_4)\bar{D} + Y_1 + Y_3D + Y_5 + Y_7$$

Logic Schematic / Circuit Diagram: Function Realization Using Single 3x8 Decoder and Gates



Part (b): K-Map Minimization [3 Marks]

$AB \setminus CD$	00	01	11	10
00	1 (m_0)	0 (m_1)	1 (m_3)	1 (m_2)
01	0 (m_4)	0 (m_5)	1 (m_7)	0 (m_6)
11	0 (m_{12})	0 (m_{13})	1 (m_{15})	1 (m_{14})
10	1 (m_8)	0 (m_9)	1 (m_{11})	1 (m_{10})

1. Quad 1 (Four Corners): $(m_0, m_2, m_8, m_{10}) \Rightarrow \bar{B}\bar{D}$.
2. Quad 2 (Column 11): $(m_3, m_7, m_{15}, m_{11}) \Rightarrow CD$.
3. Quad 3 (Rows 11 and 10, Cols 11 and 10): $(m_{15}, m_{14}, m_{11}, m_{10}) \Rightarrow AC$.

Final Verified Result

$$X(A, B, C, D) = \bar{B}\bar{D} + CD + AC$$

3.0.5 Question 5: Synchronous 2-Bit UP/DOWN Counter Schematic [7 Marks]

Problem Statement (2082 Baishakh, Q5)

Design a synchronous 2-bit up/down counter using T flip-flops.

Let mode control input be M :

- $M = 0 \Rightarrow$ UP Counting Sequence: $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$.

- $M = 1 \implies$ DOWN Counting Sequence: $00 \rightarrow 11 \rightarrow 10 \rightarrow 01 \rightarrow 00$.

1. State Transition and T Excitation Table ($T = Q_n \oplus Q_{n+1}$):

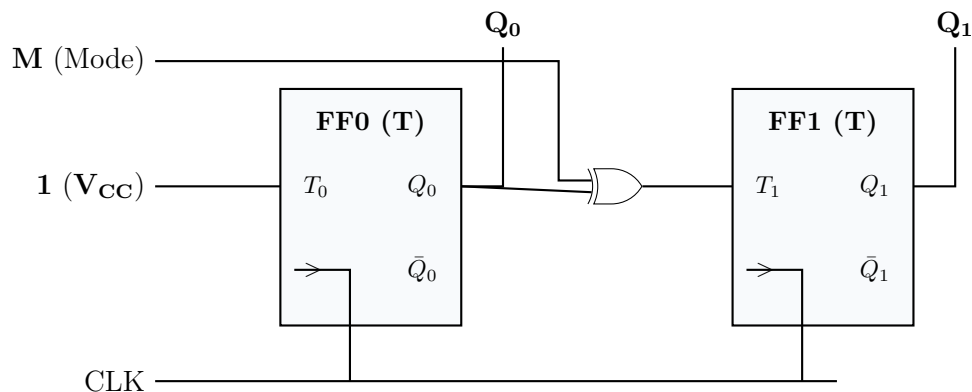
Mode M	Present State $Q_1 \quad Q_0$		Next State $Q_1^+ \quad Q_0^+$		T Inputs $T_1 \quad T_0$	
0	0	0	0	1	0	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	0	0	1	1
1	0	0	1	1	1	1
1	0	1	0	0	0	1
1	1	0	0	1	1	1
1	1	1	1	0	0	1

2. Minimized T Equations via K-Maps:

- $T_0 = 1$ (LSB flip-flop toggles on every single clock edge regardless of mode).
- For $T_1(M, Q_1, Q_0)$:

$$T_1 = \bar{M}Q_0 + M\bar{Q}_0 = M \oplus Q_0$$

Logic Schematic / Circuit Diagram: Synchronous 2-Bit UP/DOWN Counter Circuit Schematic



Final Verified Result

$$T_0 = 1, \quad T_1 = \bar{M}Q_0 + M\bar{Q}_0 = M \oplus Q_0$$

3.0.6 Question 6: 4-Bit PISO Shift Register Schematic [3 + 2 = 5 Marks]

Problem Statement (2082 Baishakh, Q6)

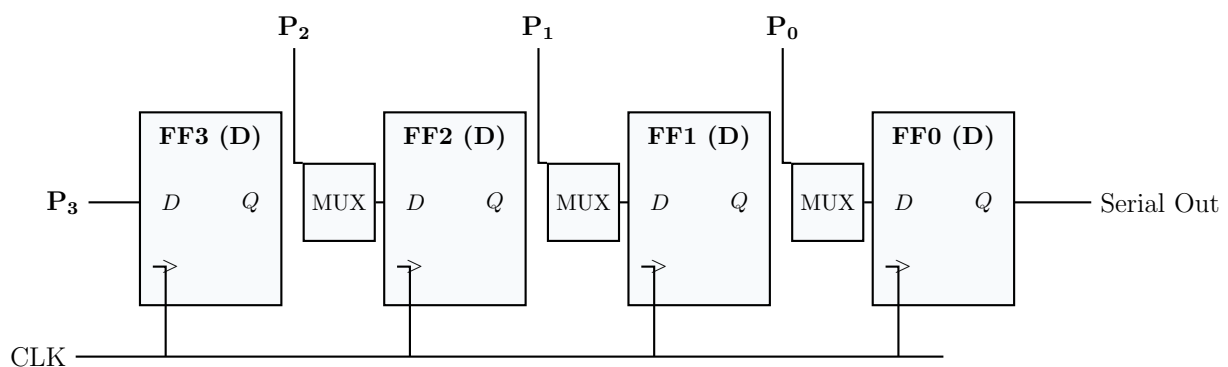
Explain the operation of 4-bit parallel-in serial-out (PISO) shift register with necessary circuit and timing diagram for 1101_2 input data.

A **Parallel-In Serial-Out (PISO)** shift register loads 4 bits of parallel data simultaneously on a single clock pulse when the mode control line $\text{Shift}/\overline{\text{Load}} = 0$, and shifts data out serially bit-by-bit when $\text{Shift}/\overline{\text{Load}} = 1$.

Steering Logic Equation for Flip-Flop Data Inputs:

$$D_i = (\text{Shift} \cdot Q_{i+1}) + (\overline{\text{Load}} \cdot \text{Parallel In}_i)$$

Logic Schematic / Circuit Diagram: 4-Bit Parallel-In Serial-Out (PISO) Shift Register



Cycle-by-Cycle Shifting of Data 1101_2 ($P_3 = 1, P_2 = 1, P_1 = 0, P_0 = 1$):

Clock Cycle	Shift/ $\overline{\text{Load}}$	Q_3	Q_2	Q_1	Q_0	Serial Output
Parallel Load	0	1	1	0	1	1 (Bit 0 available immediately)
Shift 1	1	0	1	1	0	0 (Bit 1)
Shift 2	1	0	0	1	1	1 (Bit 2)
Shift 3	1	0	0	0	1	1 (Bit 3)

Final Verified Result

The data word 1101_2 is loaded in parallel on $\text{Shift}/\overline{\text{Load}} = 0$ and shifted out serially as bits $1 \rightarrow 0 \rightarrow 1 \rightarrow 1$ across 3 consecutive shift clock cycles.

3.0.7 Question 7: Mod-12 Asynchronous Counter Schematic [3 + 3 = 6 Marks]

Problem Statement (2082 Baishakh, Q7)

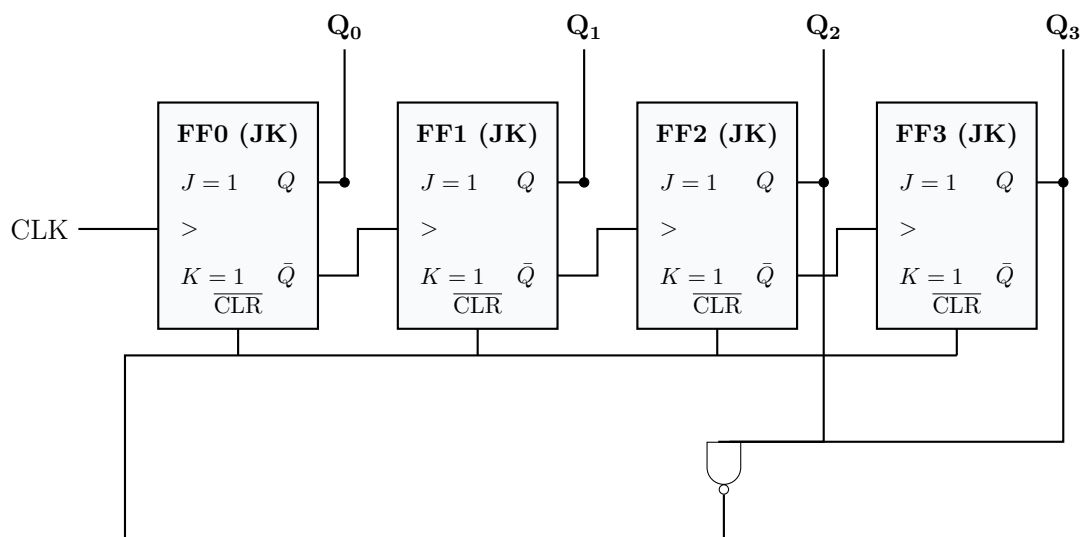
Sketch the circuit diagram of mod-12 asynchronous counter having positive-edge triggering clock system implementing JK flip-flops with neat timing diagram.

A **Mod-12 Asynchronous Counter** counts 12 states from 0000_2 (0_{10}) to 1011_2 (11_{10}). On the 12th count (1100_2 , where $Q_3 = 1, Q_2 = 1$), an asynchronous active-low clear pulse resets all flip-flops to 0000_2 :

$$\overline{\text{CLR}} = \overline{Q_3} \cdot \overline{Q_2}$$

With positive-edge triggered flip-flops, UP counting requires ripple clocking via complementary outputs: $\text{CLK}_{i+1} = \bar{Q}_i$.

Logic Schematic / Circuit Diagram: Mod-12 Positive-Edge Triggered Asynchronous Counter



Final Verified Result

Reset condition: $\overline{\text{CLR}} = \overline{Q_3} \cdot \overline{Q_2}$, providing exactly 12 stable counting states from 0000_2 to 1011_2 .

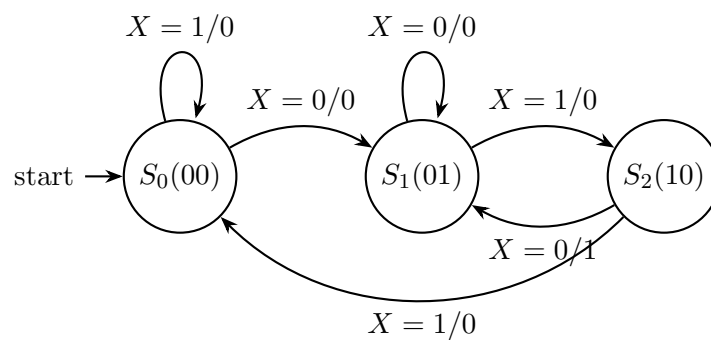
3.0.8 Question 8: FSM Sequence Detector '010' Using D Flip-Flops [10 Marks]

Problem Statement (2082 Baishakh, Q8)

Design a sequential machine that consists of one input X and one output Z . The machine is required to give output high ($Z = 1$) whenever it detects the serial sequence of '010' from its input data stream X . Implement only D flip-flops for the designed circuit realization.

1. State Definitions (Mealy Model with Overlapping Detection):

- $S_0(00)$: Reset state / No matching prefix.
- $S_1(01)$: Detected pattern '0'.
- $S_2(10)$: Detected pattern '01'.
- When in S_2 and next bit is $X = 0 \Rightarrow$ '010' detected $\Rightarrow Z = 1$, next state S_1 (since trailing '0' can start the next sequence).

Logic Schematic / Circuit Diagram: State Diagram for '010' Sequence Detector**2. State Transition and D Excitation Table ($D_1 = Q_1^+$, $D_0 = Q_0^+$):**

Present State		Input X	Next State		Output Z	D Inputs	
Q_1	Q_0		Q_1^+	Q_0^+		D_1	D_0
0	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0
0	1	0	0	1	0	0	1
0	1	1	1	0	0	1	0
1	0	0	0	1	1	0	1
1	0	1	0	0	0	0	0
1	1	0	X	X	X	X	X
1	1	1	X	X	X	X	X

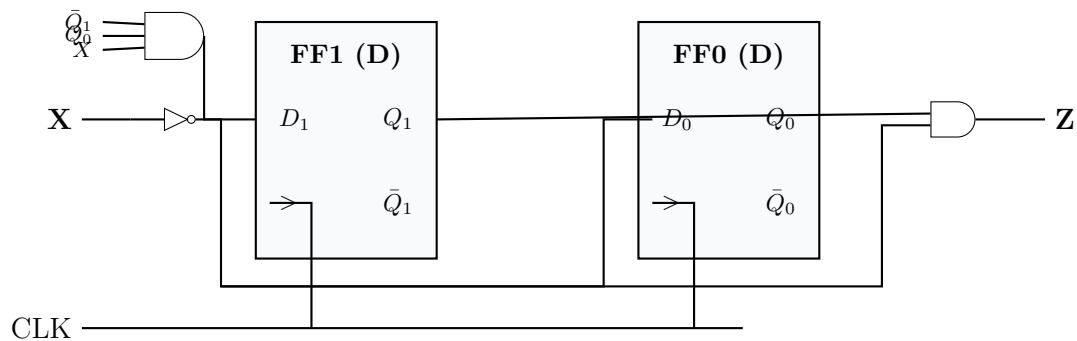
3. Minimized Equations via K-Maps:

$$D_1 = \bar{Q}_1 Q_0 X$$

$$D_0 = \bar{X}$$

$$Z = Q_1 \bar{X}$$

Logic Schematic / Circuit Diagram: Complete Logic Circuit Realization Using D Flip-Flops



Final Verified Result

$$D_1 = \bar{Q}_1 Q_0 X, \quad D_0 = \bar{X}, \quad Z = Q_1 \bar{X}$$

3.0.9 Question 9: Two-Input CMOS NAND Gate Transistor Circuit [4 + 2 = 6 Marks]

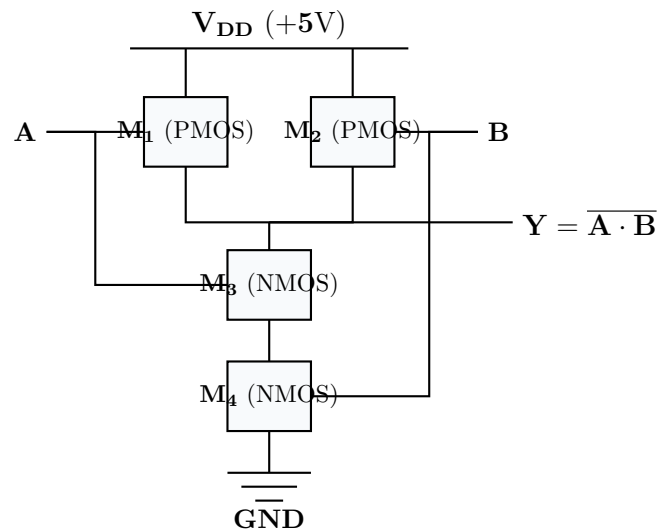
Problem Statement (2082 Baishakh, Q9)

Draw the circuit diagram of two-input CMOS NAND gate and explain its logic operation briefly and list the characteristics of TTL logic family.

Part (a): 2-Input CMOS NAND Gate Circuit and Operation [4 Marks] A CMOS logic gate consists of a **Pull-Up Network (PUN)** made of PMOS transistors and a **Pull-Down Network (PDN)** made of NMOS transistors:

- **PUN:** Two PMOS transistors (M_1, M_2) connected in **parallel** between V_{DD} and Output Y .
- **PDN:** Two NMOS transistors (M_3, M_4) connected in **series** between Output Y and Ground.

Logic Schematic / Circuit Diagram: Transistor-Level Schematic of 2-Input CMOS NAND Gate



Operation Truth Table:

A	B	M_1 (P)	M_2 (P)	M_3 (N)	M_4 (N)	Y	State
0	0	ON	ON	OFF	OFF	1 (V_{DD})	PUN conducts to V_{DD}
0	1	ON	OFF	OFF	ON	1 (V_{DD})	M_1 pulls output HIGH
1	0	OFF	ON	ON	OFF	1 (V_{DD})	M_2 pulls output HIGH
1	1	OFF	OFF	ON	ON	0 (GND)	PDN pulls output to Ground

Part (b): Characteristics of TTL Logic Family [2 Marks]

- Supply Voltage:** Standard $V_{CC} = +5.0V \pm 5\%$.
- Power Dissipation:** ≈ 10 mW per gate (Standard 7400 TTL).
- Propagation Delay:** ≈ 10 ns (high speed bipolar technology).
- Noise Margin:** $NM_H = 0.4V, NM_L = 0.4V$.
- Fan-Out:** 10 standard 7400 TTL loads.

Final Verified Result

CMOS NAND uses parallel PMOS pull-up and series NMOS pull-down transistors, providing near-zero static power dissipation and full rail-to-rail voltage swing.

3.0.10 Question 10: Time Interval Measurement Block Diagram [4 Marks]

Problem Statement (2082 Baishakh, Q10)

With the help of functional diagram explain the operation of time measuring circuit.

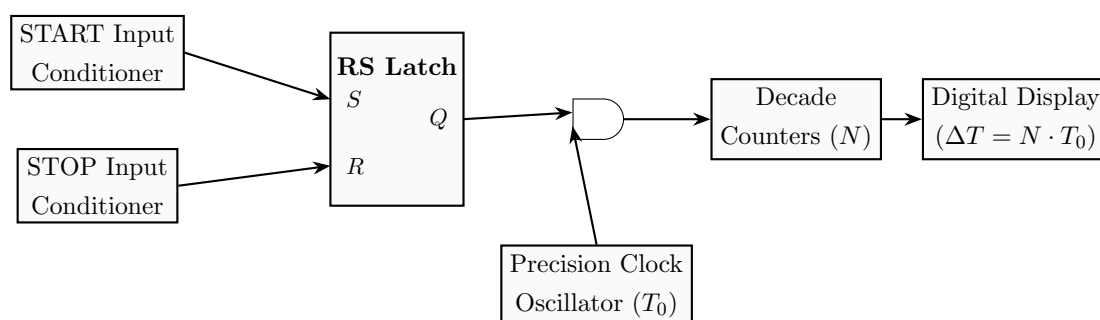
A **Digital Time Interval Meter** measures the elapsed time ΔT between a START trigger event and a STOP trigger event by counting the number of precision clock pulses of known period T_0 :

$$\Delta T = N \times T_0 = \frac{N}{f_{osc}}$$

Operational Sequence:

1. **Start Pulse Conditioning:** The START signal passes through an input amplifier and Schmitt trigger, setting the **Gate Control Flip-Flop (RS Latch)**.
2. **Gating Period:** While the latch is SET ($Q = 1$), the **Main AND Gate** opens, allowing clock pulses from the crystal oscillator time base to flow into the decade counters.
3. **Stop Pulse Conditioning:** The arrival of the STOP event resets the latch ($Q = 0$), closing the AND gate and freezing the count.
4. **Display:** The accumulated count N is displayed directly on 7-segment digital readouts in microseconds (μs), milliseconds (ms), or seconds (s).

Logic Schematic / Circuit Diagram: Functional Block Diagram of Digital Time Interval Measurement Circuit



Final Verified Result

The time interval measurement system gates standard clock pulses between START and STOP triggers, computing time elapsed as $\Delta T = N \times T_0$.

Chapter 4

2081 Ashwin Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)	Full Marks: 60
Programme: BEI, BCT	Pass Marks: 24
Year / Part: I / II	Time: 3 Hours
Subject: Digital Logic (ENEX 152)	Examination Type: Regular (New Course - 2080 Batch)

4.0.1 Question 1: Merits & Demerits of Digital Signals [2 Marks]

Problem Statement (2081 Ashwin, Q1)

Mention merits and demerits of digital signal over analog signal.

Merits of Digital Signals:

1. **Superior Noise Immunity:** Digital circuits distinguish only discrete voltage bands ('0' and '1'). As long as noise voltage does not exceed the logic threshold, signal integrity is perfectly regenerated by repeaters.
2. **Lossless Storage and Reproduction:** Data stored in semiconductor memory or optical media can be retrieved and duplicated indefinitely without analog degradation.
3. **Advanced Error Detection and Correction:** Robust mathematical codes (Parity, CRC, Hamming, Reed-Solomon) can detect and correct bit errors caused during transmission.
4. **Programmability and Integration:** Standard digital VLSI architectures (FPGA, DSP, Micro-processors) allow flexible software reconfigurability and massive circuit density.

Demerits of Digital Signals:

1. **Higher Bandwidth Requirement:** Transmitting digital square pulses requires significantly wider channel bandwidth than transmitting the raw baseband analog waveform.
2. **Quantization Error:** Analog-to-Digital Conversion (ADC) introduces unavoidable quantization noise due to finite bit resolution.
3. **System Complexity:** Requires ADC and DAC conversion stages at real-world physical interfaces.

Final Verified Result

Digital signals provide unmatched noise immunity, lossless storage, and error correction at the expense of higher transmission bandwidth and quantization artifacts.

4.0.2 Question 2: Radix & Gray Code Conversions [1.5 + 1.5 = 3 Marks]

Problem Statement (2081 Ashwin, Q2)

Perform the following as indicated:

- (a) $(6E.2C)_{16} = (?)_8$
- (b) $(10110110)_{\text{Gray}} = (?)_2$

Part (a): Hexadecimal to Octal Conversion: $(6E.2C)_{16} \rightarrow (?)_8$ [1.5 Marks]

1. **Convert Hexadecimal to 4-bit Binary:**

$$6_{16} = 0110_2, \quad E_{16} = 1110_2, \quad 2_{16} = 0010_2, \quad C_{16} = 1100_2$$

$$(6E.2C)_{16} = (0110 \ 1110 . 0010 \ 1100)_2$$

2. **Regroup into 3-bit Octal Groups:**

- **Integer part (grouping leftward from radix point):**

$$\underbrace{001}_1 \underbrace{101}_5 \underbrace{110}_6 \implies (156)_8$$

- **Fractional part (grouping rightward from radix point):**

$$\underbrace{001}_1 \underbrace{011}_3 \underbrace{000}_0 \implies (.130)_8 = (.13)_8$$

Thus, $(6E.2C)_{16} = (156.13)_8$.

Part (b): Gray Code to Binary Conversion: $(10110110)_{\text{Gray}} \rightarrow (?)_2$ [1.5 Marks] Given $g_7g_6g_5g_4g_3g_2g_1g_0 = 10110110_2$:

- $b_7 = g_7 = 1$
- $b_6 = b_7 \oplus g_6 = 1 \oplus 0 = 1$
- $b_5 = b_6 \oplus g_5 = 1 \oplus 1 = 0$
- $b_4 = b_5 \oplus g_4 = 0 \oplus 1 = 1$
- $b_3 = b_4 \oplus g_3 = 1 \oplus 0 = 1$
- $b_2 = b_3 \oplus g_2 = 1 \oplus 1 = 0$
- $b_1 = b_2 \oplus g_1 = 0 \oplus 1 = 1$
- $b_0 = b_1 \oplus g_0 = 1 \oplus 0 = 1$

Thus, $(10110110)_{\text{Gray}} = (\mathbf{11011011})_2$.

Final Verified Result

$$\begin{aligned} \text{(a)} \quad (6E.2C)_{16} &= (\mathbf{156.13})_8 \\ \text{(b)} \quad (10110110)_{\text{Gray}} &= (\mathbf{11011011})_2 \end{aligned}$$

4.0.3 Question 3: BCD to 7-Segment 'f' Segment Decoder Logic Circuit [4 Marks]

Problem Statement (2081 Ashwin, Q3)

Design the simplest logic circuit for 'f' segment for the BCD-to-seven segment display decoder.

A 7-segment display uses LED segments a, b, c, d, e, f, g . The 'f' segment represents the upper-left vertical LED segment, which must be illuminated (HIGH) for decimal digits **0, 4, 5, 6, 8, 9**.

1. Truth Table for Segment $f(A, B, C, D)$ (where A is MSB):

Decimal Digit	A	B	C	D	f	Display Appearance
0	0	0	0	0	1	Illuminated
1	0	0	0	1	0	Off
2	0	0	1	0	0	Off
3	0	0	1	1	0	Off
4	0	1	0	0	1	Illuminated
5	0	1	0	1	1	Illuminated
6	0	1	1	0	1	Illuminated
7	0	1	1	1	0	Off
8	1	0	0	0	1	Illuminated
9	1	0	0	1	1	Illuminated
10–15	Invalid BCD Codes				X	Don't Cares ($d_{10} \dots d_{15}$)

2. 4-Variable K-Map Minimization:

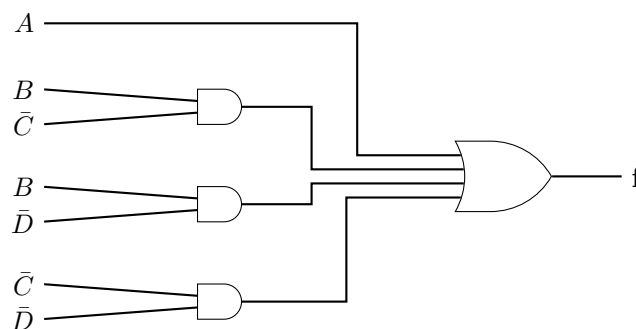
$AB \setminus CD$	00	01	11	10
00	1 (m_0)	0 (m_1)	0 (m_3)	0 (m_2)
01	1 (m_4)	1 (m_5)	0 (m_7)	1 (m_6)
11	X (d_{12})	X (d_{13})	X (d_{15})	X (d_{14})
10	1 (m_8)	1 (m_9)	X (d_{11})	X (d_{10})

Groupings:

1. Group 1 (Octet in rows 11 & 10): ($m_8, m_9, d_{10}, d_{11}, d_{12}, d_{13}, d_{14}, d_{15}$) $\Rightarrow A$.
2. Group 2 (Quad in columns 00 & 01, rows 01 & 11): (m_4, m_5, d_{12}, d_{13}) $\Rightarrow B\bar{C}$.
3. Group 3 (Quad in column 00): (m_0, m_4, d_{12}, m_8) $\Rightarrow \bar{C}\bar{D}$.
4. Group 4 (Quad across rows 01 & 11, cols 00 & 10): (m_4, m_6, d_{12}, d_{14}) $\Rightarrow B\bar{D}$.

Minimal Boolean Expression:

$$f = A + B\bar{C} + B\bar{D} + \bar{C}\bar{D}$$

Logic Schematic / Circuit Diagram: Logic Circuit Realization for Segment 'f'**Final Verified Result**

$$f = A + B\bar{C} + B\bar{D} + \bar{C}\bar{D}$$

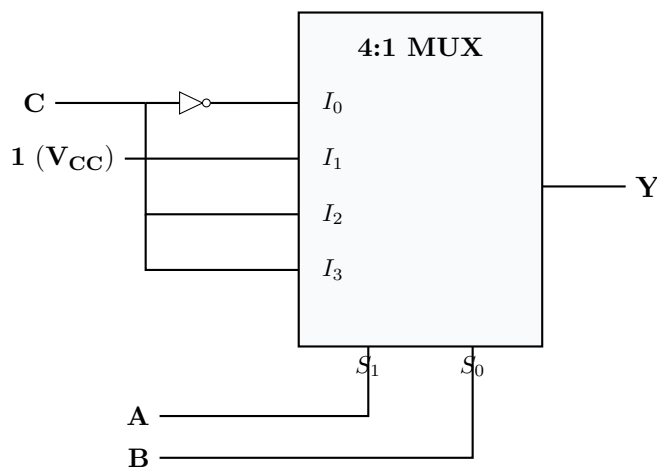
4.0.4 Question 4: 4:1 MUX Realization & Combined Adder/Subtractor [3 + 7 = 10 Marks]**Problem Statement (2081 Ashwin, Q4)**

Implement $Y(A, B, C) = \sum m(0, 2, 3, 5, 7)$ using only a single 4 : 1 MUX. Design a circuit which can realize both the half-adder and the half-subtractor in a single circuit.

Part (a): Realize $Y(A, B, C) = \sum m(0, 2, 3, 5, 7)$ Using Single 4 : 1 MUX [3 Marks] Let select lines be $S_1 = A, S_0 = B$:

Select Inputs (A, B)	Minterms Included	Function Output Y	Data Input Line
00	$m_0(\bar{C}) + m_1(C) \rightarrow m_0$	$\bar{C}$	$I_0 = \bar{C}$
01	$m_2(\bar{C}) + m_3(C) \rightarrow m_2, m_3$	$\bar{C} + C = 1$	$I_1 = 1 (V_{CC})$
10	$m_4(\bar{C}) + m_5(C) \rightarrow m_5$	C	$I_2 = C$
11	$m_6(\bar{C}) + m_7(C) \rightarrow m_7$	C	$I_3 = C$

Logic Schematic / Circuit Diagram: 4:1 Multiplexer Realization for $Y(A, B, C) = \sum m(0, 2, 3, 5, 7)$



Part (b): Combined Half-Adder and Half-Subtractor in a Single Circuit [7 Marks] Let A and B be the two 1-bit binary inputs, and let M be the mode control line:

- **Mode $M = 0$ (Half-Adder):**

$$\text{Sum } S = A \oplus B, \quad \text{Carry } C = A \cdot B$$

- **Mode $M = 1$ (Half-Subtractor):**

$$\text{Difference } D = A \oplus B, \quad \text{Borrow } B_{\text{out}} = \bar{A} \cdot B$$

Key Insight: The first output is identical in both operations: Sum/Difference = $A \oplus B$.

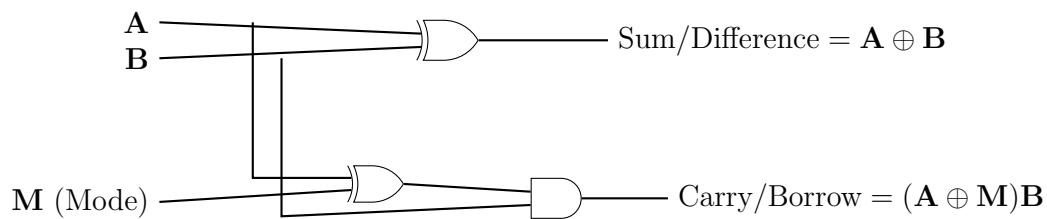
The second output (Carry/Borrow) requires $A \cdot B$ when $M = 0$ and $\bar{A} \cdot B$ when $M = 1$. Using an XOR gate with mode control M :

$$(A \oplus M) = \begin{cases} A \oplus 0 = A & \text{when } M = 0 \\ A \oplus 1 = \bar{A} & \text{when } M = 1 \end{cases}$$

Therefore, the combined Carry / Borrow output equation is:

$$\text{Carry/Borrow} = (A \oplus M) \cdot B$$

Logic Schematic / Circuit Diagram: Combined Half-Adder and Half-Subtractor Circuit Schematic



Final Verified Result

4:1 MUX Inputs: $I_0 = \bar{C}, I_1 = 1, I_2 = C, I_3 = C$ with $S_1 = A, S_0 = B$.

Combined Adder/Subtractor Equations:

$$\text{Sum/Difference} = A \oplus B, \quad \text{Carry/Borrow} = (A \oplus M) \cdot B$$

4.0.5 Question 5: Mod-6 Synchronous DOWN Counter Schematic [7 Marks]

Problem Statement (2081 Ashwin, Q5)

Design a mod-6 synchronous down counter using JK flip-flops.

A **Mod-6 Synchronous DOWN Counter** counts backward through 6 states:

$$5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 5 \quad (101_2 \rightarrow 100_2 \rightarrow 011_2 \rightarrow 010_2 \rightarrow 001_2 \rightarrow 000_2 \rightarrow 101_2)$$

Requires $N = 3$ flip-flops (Q_2, Q_1, Q_0). Unused states: 110_2 and 111_2 (treated as Don't Cares).

1. State Transition and JK Excitation Table:

State	Present State			Next State			JK Flip-Flop Inputs					
	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	J_2	K_2	J_1	K_1	J_0	K_0
5	1	0	1	1	0	0	X	0	0	X	X	1
4	1	0	0	0	1	1	X	1	1	X	1	X
3	0	1	1	0	1	0	0	X	X	0	X	1
2	0	1	0	0	0	1	0	X	X	1	1	X
1	0	0	1	0	0	0	0	X	0	X	X	1
0	0	0	0	1	0	1	1	X	0	X	1	X

2. Minimized JK Equations via K-Maps:

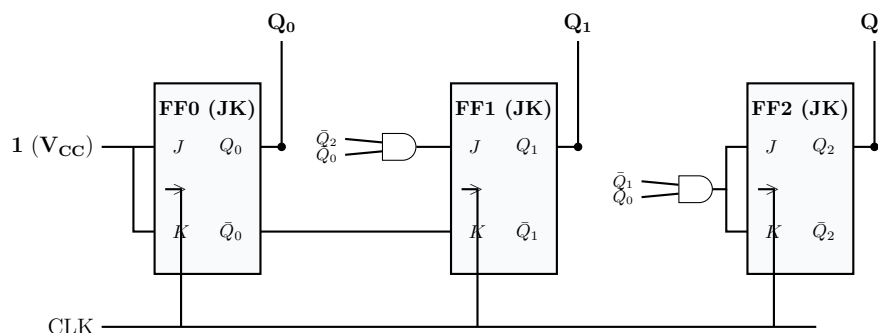
- $J_0 = 1, \quad K_0 = 1$ (LSB toggles every single clock edge).

- $J_1 = \bar{Q}_2 \bar{Q}_0$, $K_1 = \bar{Q}_0$.
- $J_2 = \bar{Q}_1 \bar{Q}_0$, $K_2 = \bar{Q}_1 \bar{Q}_0$ (or $K_2 = \bar{Q}_0$).

3. Lock-out and Self-Starting Verification:

- If counter powers up in state 110_2 : $Q_2^+ = 1, Q_1^+ = 0, Q_0^+ = 1 \Rightarrow$ Next State 101_2 (Valid state!).
- If counter powers up in state 111_2 : $Q_2^+ = 1, Q_1^+ = 1, Q_0^+ = 0 \Rightarrow$ Next State $110_2 \rightarrow 101_2$ (Valid state!).

Logic Schematic / Circuit Diagram: Mod-6 Synchronous DOWN Counter Logic Circuit



Final Verified Result

$$J_0 = 1, K_0 = 1, J_1 = \bar{Q}_2 \bar{Q}_0, K_1 = \bar{Q}_0, J_2 = K_2 = \bar{Q}_1 \bar{Q}_0$$

4.0.6 Question 7: 4-Bit SIPO Shift Register Schematic & Waveforms [5 Marks]

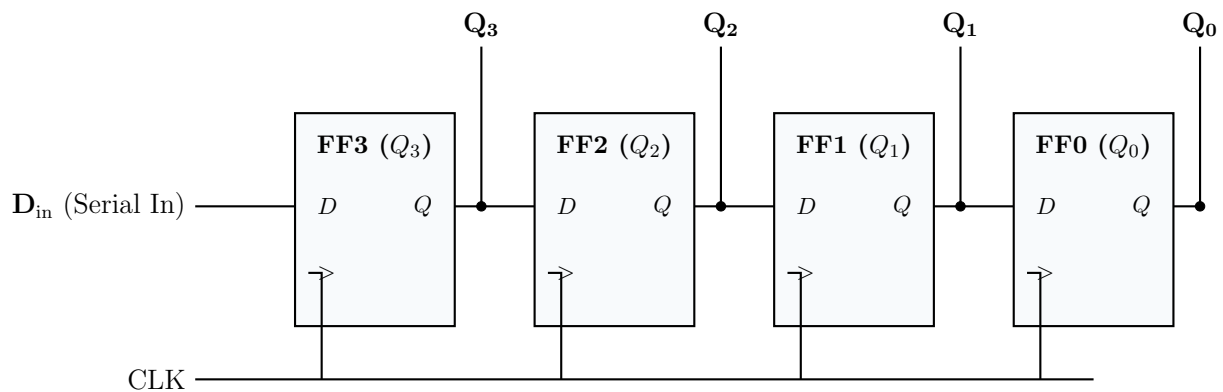
Problem Statement (2081 Ashwin, Q7)

Explain the working of 4-bit SIPO shift register with timing diagram for data input '1011'.

1. Circuit Configuration:

- 4 D flip-flops cascaded in series ($D_{in} \rightarrow D_3, Q_3 \rightarrow D_2, Q_2 \rightarrow D_1, Q_1 \rightarrow D_0$).
- Common clock shifts data one stage right on each positive clock edge.
- Parallel outputs Q_3, Q_2, Q_1, Q_0 available simultaneously after 4 pulses.

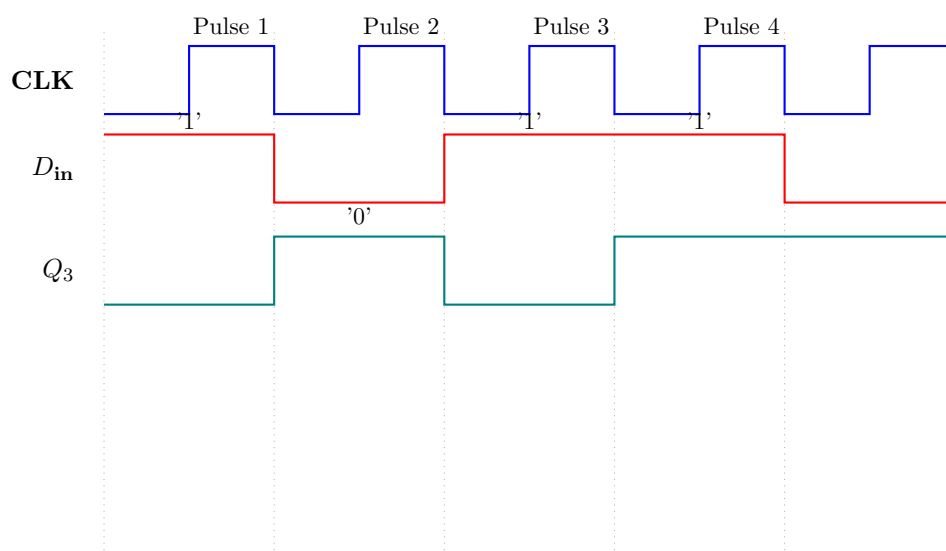
Logic Schematic / Circuit Diagram: 4-Bit SIPO Shift Register Circuit Schematic



2. Shift Sequence Table for 1011_2 (MSB entered first): Assume initial contents $Q_3Q_2Q_1Q_0 = 0000_2$.

Clock Pulse	Serial Input (D_{in})	Q_3	Q_2	Q_1	Q_0	State Summary
Initial	—	0	0	0	0	Cleared state
CLK 1	1 (MSB)	1	0	0	0	First bit into FF3
CLK 2	0	0	1	0	0	Shift right, second bit in
CLK 3	1	1	0	1	0	Shift right, third bit in
CLK 4	1 (LSB)	1	1	0	1	Complete word loaded (1011_2)

Logic Schematic / Circuit Diagram: Timing Waveforms for Shifting 1011_2 into SIPO Register



Final Verified Result

Parallel output after 4 clock pulses is $Q_3Q_2Q_1Q_0 = 1101_2$ (corresponding to serial arrival sequence 1011_2).

4.0.7 Question 8: 3-Bit Asynchronous UP/DOWN Counter Schematic [5 Marks]

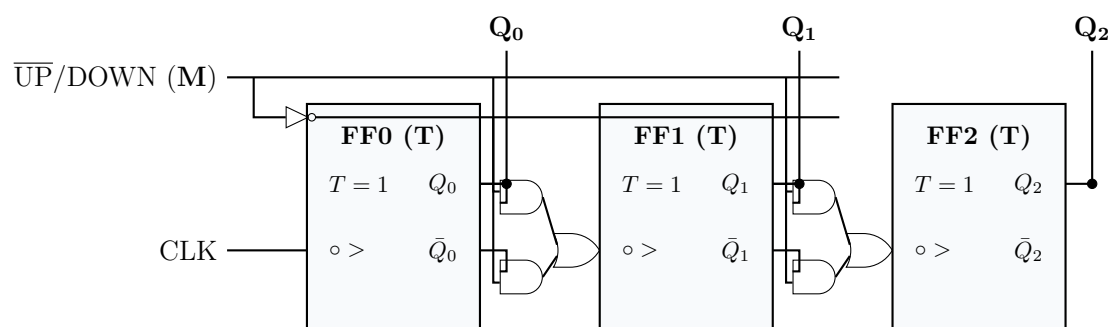
Problem Statement (2081 Ashwin, Q8)

Explain the operation of 3-bit asynchronous up/down counter with logic diagram.

An asynchronous UP/DOWN counter uses a control line M to steer either the normal output Q (for DOWN counting with negative-edge clocking) or inverted output $\bar{Q}$ (for UP counting with negative-edge clocking) to the clock input of the next stage:

$$\text{CLK}_{i+1} = \bar{M}Q_i + M\bar{Q}_i = M \oplus Q_i$$

Logic Schematic / Circuit Diagram: 3-Bit Negative-Edge Triggered Asynchronous UP/DOWN Counter



Final Verified Result

$M = 0 \implies$ UP Counting ($Q_0 \rightarrow \text{CLK}_1, Q_1 \rightarrow \text{CLK}_2$), $M = 1 \implies$ DOWN Counting ($\bar{Q}_0 \rightarrow \text{CLK}_1, \bar{Q}_1 \rightarrow \text{CLK}_2$).

4.0.8 Question 9: Synchronous FSM Sequence Detector '101' [7 Marks]

Problem Statement (2081 Ashwin, Q9)

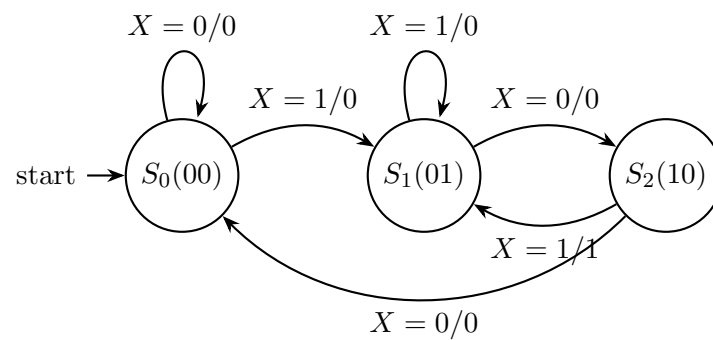
Design a synchronous sequential machine that has 1-bit serial input X and output Z which will be high when the input contains the message '101' (Use SR Flip-Flop).

1. State Definitions and State Transition Diagram (Mealy Model):

- $S_0(00)$: Reset state.
- $S_1(01)$: Sequence '1' detected.
- $S_2(10)$: Sequence '10' detected.

- In S_2 , when $X = 1 \Rightarrow$ '101' detected $\Rightarrow Z = 1$, next state S_1 (overlapping allowed).

Logic Schematic / Circuit Diagram: State Transition Diagram for '101' Sequence Detector



2. State and Excitation Table for SR Flip-Flops (A, B):

Present State		Input X	Next State		Output Z	SR Inputs			
A	B		A^+	B^+		S_A	R_A	S_B	R_B
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	0	X	1	0
0	1	0	1	0	0	1	0	0	1
0	1	1	0	1	0	0	X	X	0
1	0	0	0	0	0	0	1	0	X
1	0	1	0	1	1	0	1	1	0
1	1	0	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X

3. Minimized Equations via K-Maps:

$$S_A = B\bar{X}$$

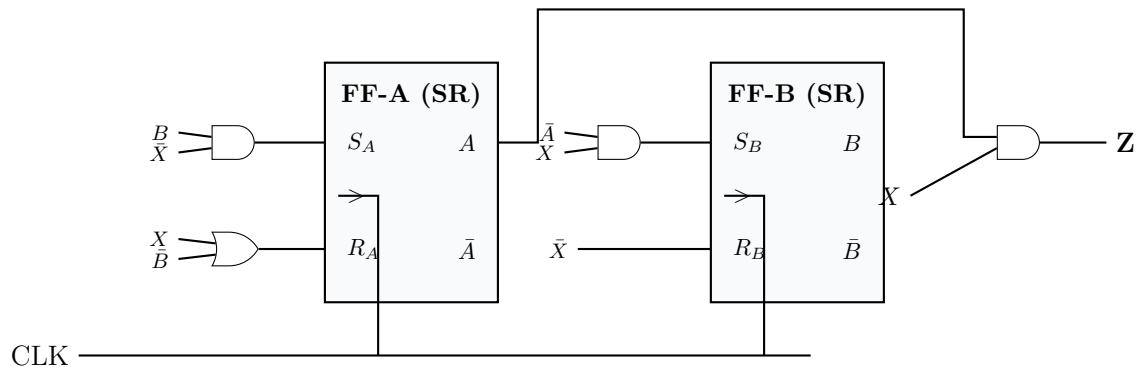
$$R_A = X + \bar{B}$$

$$S_B = \bar{A}X$$

$$R_B = \bar{X}$$

$$Z = AX$$

Logic Schematic / Circuit Diagram: Complete Synchronous Logic Schematic Using SR Flip-Flops



Final Verified Result

$$S_A = B\bar{X}, \quad R_A = X + \bar{B}, \quad S_B = \bar{A}X, \quad R_B = \bar{X}, \quad Z = AX$$

4.0.9 Question 9: Two-Input TTL NOR Gate Transistor Circuit [5 + 2 = 7 Marks]

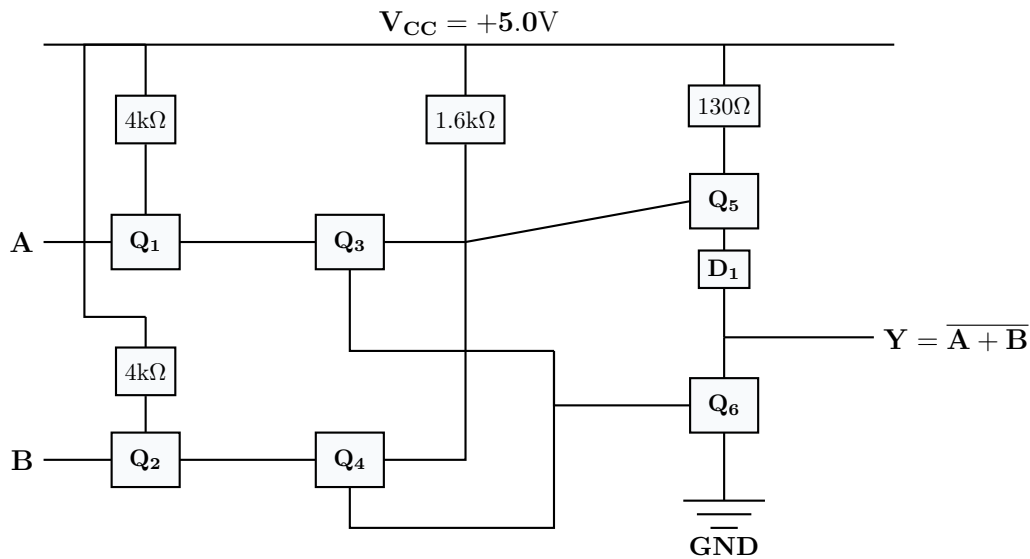
Problem Statement (2081 Ashwin, Q9)

Draw the circuit diagram of two-input TTL NOR gate and explain its logic operation briefly and list the characteristics of CMOS logic family.

Part (a): 2-Input TTL NOR Gate Transistor Circuit & Operation [5 Marks] A Transistor-Transistor Logic (TTL) NOR gate uses:

- **Dual Input Stages (Parallel branches):**
 - Input A drives input transistor Q_1 (with pull-up $R_1 = 4\text{k}\Omega$) which controls phase-splitter Q_3 .
 - Input B drives input transistor Q_2 (with pull-up $R_2 = 4\text{k}\Omega$) which controls phase-splitter Q_4 .
- **Parallel Phase Splitters ($Q_3 \parallel Q_4$):** Q_3 and Q_4 share a common collector load ($R_3 = 1.6\text{k}\Omega$) and a common emitter resistor ($R_4 = 1\text{k}\Omega$).
- **Output Totem-Pole Stage:** Upper active pull-up transistor Q_5 with series diode D_1 and current-limiting resistor $R_5 = 130\Omega$, and lower pull-down transistor Q_6 .

Logic Schematic / Circuit Diagram: Transistor Schematic of 2-Input Standard TTL NOR Gate



Logic Operation Table:

A	B	Q ₁	Q ₂	Q ₃	Q ₄	Q ₅	Q ₆	Y	Logic State
0	0	Sat	Sat	OFF	OFF	ON	OFF	1 ($\approx 3.4V$)	High Output (V_{OH})
0	1	Sat	Inv	OFF	ON	OFF	ON	0 ($\approx 0.2V$)	Low Output (V_{OL})
1	0	Inv	Sat	ON	OFF	OFF	ON	0 ($\approx 0.2V$)	Low Output (V_{OL})
1	1	Inv	Inv	ON	ON	OFF	ON	0 ($\approx 0.2V$)	Low Output (V_{OL})

Part (b): Characteristics of CMOS Logic Family [2 Marks]

- Extremely Low Static Power Consumption:** Near zero (< 10 nW per gate), power is consumed only during switching transitions.
- Wide Supply Voltage Range:** Operates reliably from +3V to +15V (4000 series) or +2V to +6V (74HC series).
- Very High Noise Immunity:** Noise margin is typically 40% to 45% of V_{DD} ($> 1.5V$ at $V_{DD} = 5V$).
- Extremely High Fan-Out:** Fan-out > 50 due to nearly infinite MOSFET input resistance ($R_{in} > 10^{12} \Omega$).
- Full Rail-to-Rail Output Swing:** $V_{OH} \approx V_{DD}$ and $V_{OL} \approx 0V$.

Final Verified Result

The TTL NOR gate uses parallel phase splitters (Q_3, Q_4) to pull the totem-pole output LOW whenever either input is HIGH. CMOS logic provides near-zero static power dissipation and high noise margins.

4.0.10 Question 10: Digital Frequency Measurement Block Diagram [4 Marks]

Problem Statement (2081 Ashwin, Q10)

With the help of functional diagram explain the operation of frequency measurement.

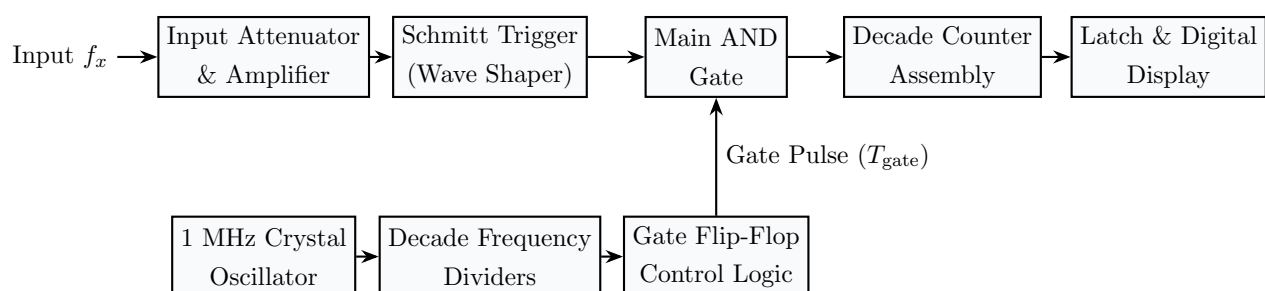
A **Digital Frequency Measurement System** determines the unknown frequency f_x of an input waveform by accumulating the count of input cycles during a calibrated time gate interval T_{gate} :

$$f_x = \frac{N}{T_{\text{gate}}}$$

Operational Principle:

- 1. Input Waveform Conditioning:** The unknown periodic analog signal is amplified and passed through a **Schmitt Trigger**, converting it into a clean train of digital clock pulses of frequency f_x .
- 2. Time Base Generation:** A high-precision crystal oscillator (e.g. 1 MHz, stability ± 1 ppm) feeds a chain of decade frequency dividers to synthesize accurate gate open times $T_{\text{gate}} = 1$ s, 0.1 s, 10 ms.
- 3. Gating Logic:** The Main AND Gate is enabled by the gate control bistable multivibrator for exactly T_{gate} . Exactly $N = f_x \times T_{\text{gate}}$ pulses pass through the gate.
- 4. Decade Counting & Readout:** The cascaded BCD counter counts N pulses, transfers the result to a latch register, and illuminates the 7-segment digital display.

Logic Schematic / Circuit Diagram: Block Diagram of Digital Frequency Measurement System



Final Verified Result

Frequency is measured by counting N conditioned input pulses over a precision gate window T_{gate} via the formula $f_x = N/T_{\text{gate}}$.